

"DIGITAL ELECTRONICS"



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1. Digital fundamentals, Thomas Floyd, Pearson Pys.
2. Digital Design, (Morris Mano), Pearson.
3. Fundamental of Digital circuit (A. Anand Kumar), PHI Publication.

Electronics system are classified into two categories.

generally ~~ref~~ referred to as analog and digital.

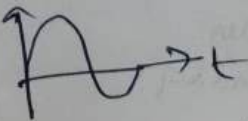
generally there are two types of signal analog signal and digital signal.

A continuously varying signal (V or I) is called an analog signal for exp., a sinusoidal signal is a analog signal however, a signal (V or I) which can have only two discrete value is called digital signal.

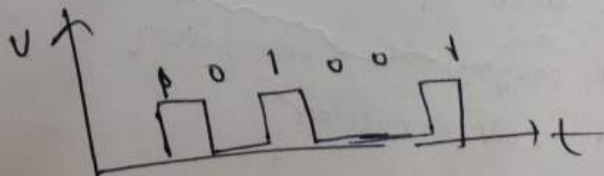
Analog and Digital system

System Basically a system is combination of device designed to perform a specific fun. System classified into two major categories. ~~analog~~ analog sy and digital system.

Analog system - Analog system is a system which processes analog signal.



Digital system Digital system is a system, which processes discrete or digital signal. It is a combination of device designed to perform logical information that are represented in digital form.



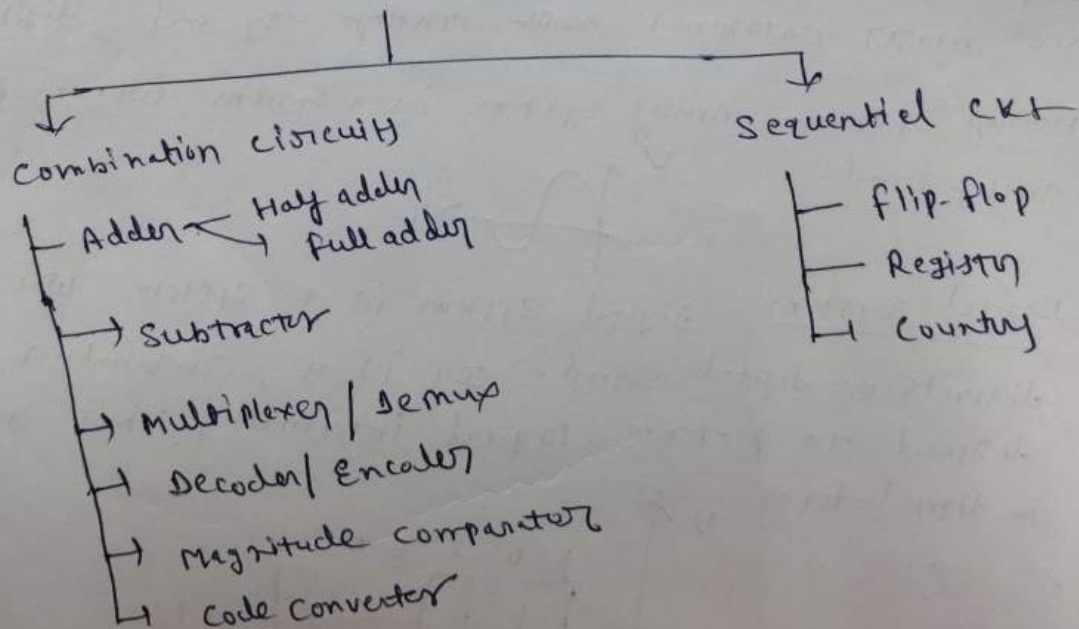
Advantage of Digital system

1. Digital system are generally easier to design. any digital sy can be constructed by using switching ckt. The switching circuit have two Value Low or High.
2. Accuracy and Precision are greater as compared to the analog system.

Digital electronics

- logical expression specification $\rightarrow$ boolean Algebra
K-map
- logic gate
- Number System

Digital circuit



Logic gates:

Logic gate is a digital ckt that is able to operate on a number of binary inputs in order to perform a particular logic function. The gate is a digital ckt with one or more inputs, but only one output.

basic logic gates

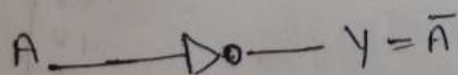
1. NOT
2. AND
3. OR

universal gates

4. NAND
5. NOR
6. XOR } Exclusive gate
7. XNOR }

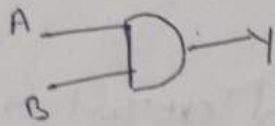
Truth table →

1. Not gate



inp	oup
A	Y
0	1
1	0

2. AND gate

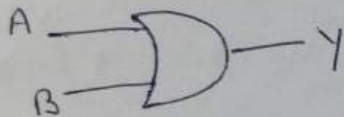


$$Y = A \cdot B$$

Truth table

i/p		o/p
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

3. OR gate



$$Y = A + B$$

Truth table

i/p		o/p
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

universal gate →

NAND & NOR are universal gate. we can obtain any logic expression using only NAND or only NOR.
So These gate are called universal gate.

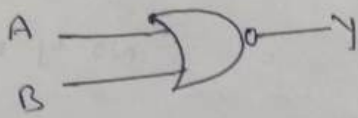
4. NAND →



$$Y = \overline{A \cdot B}$$

i/p		o/p
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

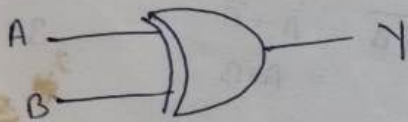
5. NOR →



$$Y = \overline{A+B}$$

inp		o/p
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6. Ex-OR gate —
Exclusive OR gate



$$Y = A \oplus B$$

$$Y = A\bar{B} + \bar{A}B$$

inp		o/p
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7. EX-NOR gate



$$Y = A \odot B$$

$$Y = AB + \bar{A}\bar{B}$$

inp		o/p
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Digital electronics →

NAND & NOR are universal gate

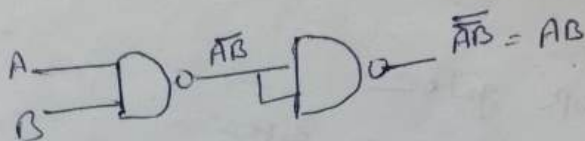
No. of NAND gate

NAND → NOT



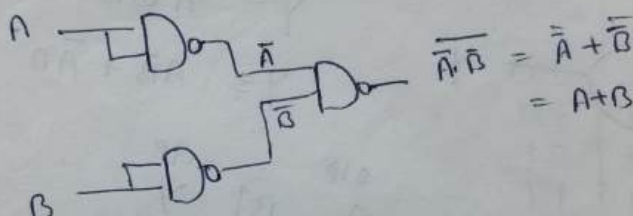
1

AND ⇒



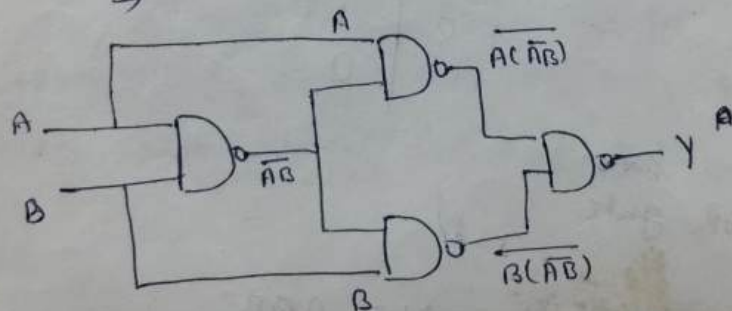
2

OR ⇒



3

EXOR ⇒



= 4

$$Y = \overline{\overline{A(\bar{A}\bar{B})} \cdot \overline{B(\bar{A}\bar{B})}}$$

$$= \overline{A(\bar{A}\bar{B})} + \overline{B(\bar{A}\bar{B})}$$

$$= A(\bar{A}\bar{B}) + B(\bar{A}\bar{B})$$

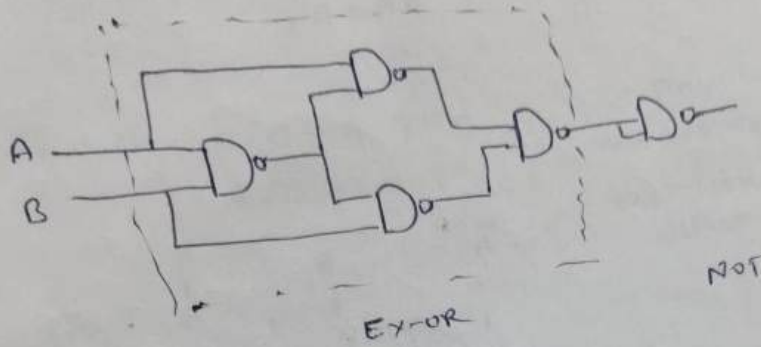
$$= A(\bar{A} + \bar{B}) + B(\bar{A} + \bar{B})$$

$$= \underset{0}{A\bar{A}} + A\bar{B} + \underset{0}{B\bar{A}} + B\bar{B}$$

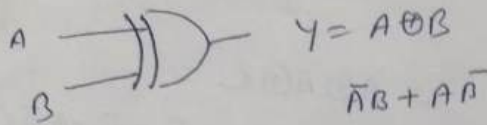
$$= A\bar{B} + \bar{A}B = A \oplus B$$

N. of NAND gate

ExNOR $\Rightarrow$



Ex-OR - gate (XOR gate)

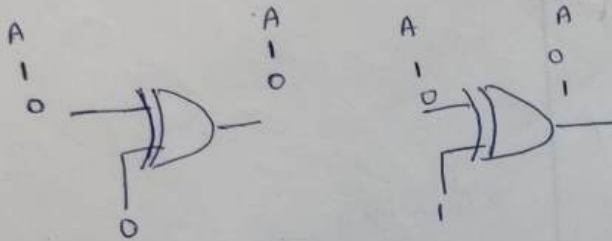


XOR gate

$$A = B \Rightarrow Y = 0$$

$$A \neq B \Rightarrow Y = 1$$

It produces as O/P 1 when the I/P are not equal, it is called anti-coincidence gate or inequality detector.



O/P change as I/P change s/p
act as buffer

act as Not gate

$$A \oplus A = 0$$

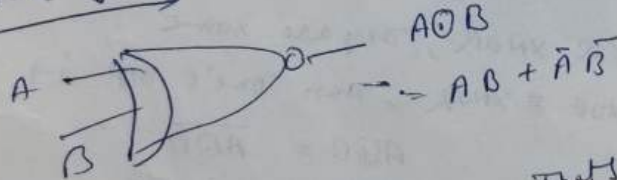
$$A \oplus A \oplus A = A$$

$$A \oplus A \oplus A \oplus A = 0$$

when odd no. of same variable are XOR, then O/P is A

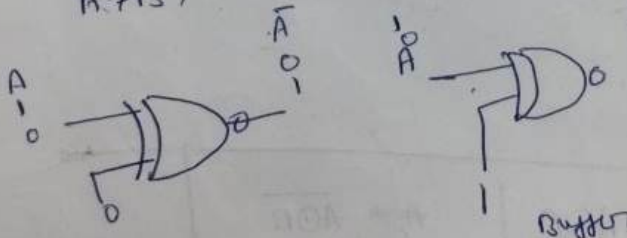
when even no. of same var. are XOR, then O/P = 0

XNOR gate



when $A = B$, O/P is 1
 $A \neq B$, O/P is 0

That's why known as Equality detector or coincidence gate



Nor gate.

Buffer

If, one I/P is complement in XOR we get XNOR & vice-versa.

$$\bar{A} \oplus B = A \odot B$$

$$\bar{A} \odot B = A \oplus B$$

$$A \oplus A = 1$$

$$A \oplus A \oplus A = A$$

$$A \oplus A \oplus A \oplus A = 1$$

Sum = 10
diff =

A	B	C	$A \oplus B \oplus C$	$A \oplus B \oplus C$
0	0	0	0	0
0	0	1	1	1
0	1	0	1	
0	1	1	0	
1	0	0	1	
1	0	1	0	
1	1	0	0	
1	1	1	1	

$$A \oplus B \oplus C =$$

$$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

When odd no. of 1's are given. Then O/P is 1
 When odd N. of 1's in XOR & XNOR, they are same
 When even no. of 1's in XOR & XNOR, then one's O/P is 1

$$A \oplus B = \overline{A \odot B}$$

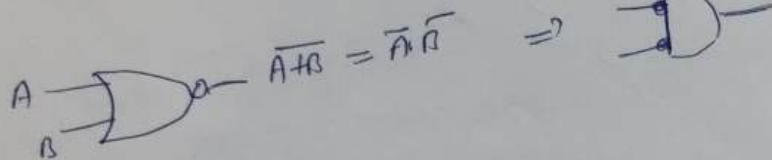
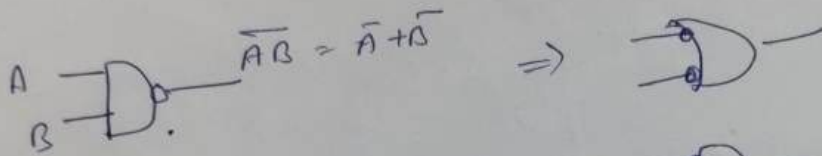
$$A \oplus B \oplus C = \overline{A \odot B \odot C}$$

$$A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$$

~~$$A \oplus B = \overline{A \odot B}$$~~
~~$$A \oplus B \oplus C = \overline{A \odot B \odot C}$$~~
~~$$A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$$~~

A	B	$A \oplus B$	$A \odot B$	$A \oplus B$
0	0	0	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	1	0

Logic gate	No. of NAND	No. of NOR
NOT	1	1
AND	2	3
OR	3	2
EXOR	4	5
EXNOR	5	4



Boolean Algebra

Boolean algebra defines constants, Variable & function to describe binary system. It then describes a number of Theorems that can be used to manipulate logic expressions.

NOT $\bar{A} = A$

AND

$$A \cdot A = A$$

$$A \cdot 1 = A$$

$$A \cdot 0 = 0$$

$$A \cdot \bar{A} = 0$$

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

OR

$$A + A = A$$

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + \bar{A} = 1$$

Ex.

$$AB + A\bar{B}$$

$$A(B + \bar{B})$$

$$= A \cdot 1$$

$$= A$$

Ex.

$$AB + A\bar{B}C + A\bar{B}\bar{C}$$

$$AB + A\bar{B}(C + \bar{C})$$

$$AB + A\bar{B} \cdot 1$$

$$AB + A\bar{B}$$

$$A(B + \bar{B})$$

$$A \cdot 1$$

$$A$$

$$Q. (A+B)(A+C)$$

$$A(A+C) + B(A+C)$$

$$AA + AC + BA + BC$$

$$A + AC + AB + BC$$

$$A(1+C) + AB + BC$$

$$A + AB + BC$$

$$A(1+B) + BC$$

$$A + BC$$

De Morgan's Theorem

Theorem I - The complement of a product is equal to the sum of the complements.

$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

Theorem II - The complement of a sum is equal to the product of the complements.

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

Q

$$AB + A(B+C) + B(B+C)$$

$$AB + AB + AC + BB + BC$$

$$AB + AC + B + BC$$

$$AB + AC + B(1+C)$$

$$AB + AC + B$$

$$B(A+1) + AC$$

$$\underline{B + AC}$$

Q.

Simplify

$$\overline{\overline{A+B\bar{C}} + D \cdot (E+\bar{F})}$$

$$\overline{\overline{A+B\bar{C}}} \cdot \overline{D \cdot (E+\bar{F})}$$

$$A+B\bar{C} \cdot (\bar{D} + \overline{E+\bar{F}})$$

$$(A+B\bar{C}) (\bar{D} + E + \bar{F}) \text{ Ans}$$

Q.

$$(\bar{A}+B) (A+B+D) \bar{D}$$

$$(\bar{A}+B) (A\bar{D} + B\bar{D} + D\bar{D})$$

$$(\bar{A}+B) (A\bar{D} + B\bar{D} + 0)$$

$$(\bar{A}+B) (A\bar{D} + B\bar{D})$$

$$\bar{A}A\bar{D} + \bar{A}B\bar{D} + A\bar{D}B + B\bar{D}B$$

$$0 + \bar{A}B\bar{D} + A\bar{D}B + B\bar{D}$$

$$B\bar{D} (A+\bar{A}) + B\bar{D}$$

$$B\bar{D} (1) + B\bar{D}$$

$$B\bar{D} \text{ Ans}$$

$$(\bar{A}+B) (A+B+D) \bar{D}$$

Boolean Function → The boolean variable are either in complemented or in uncomplemented form & the term are arranged in one of the two standard form of boolean fun.

1. Sum of Product form SOP
2. Product of Sum form POS

1. SOP → $AB\bar{C} + \bar{B}C + AB$

2. POS $(B+C) \cdot (A+B+\bar{C}) \cdot (A+\bar{B})$

Canonical Logic form —

We can realize that all the individual terms don't involve all the literals (variables).

If each term in SOP & POS forms containing all the variable then these are known as canonical (standard) SOP & POS expression.

$$Y = \cancel{A\bar{C}} + AB + BC$$

$$Y = A + \bar{A}B \quad \text{change in canonical}$$

$$Y = A(B + \bar{B}) + \bar{A}B$$

$$= AB + A\bar{B} + \bar{A}B$$

This is standard SOP form or canonical form.

$$Y = (A+B)(B+C)(C+A)$$

$$= (A+B+C)(B+C+A\bar{A})(C+A+B\bar{B})$$

$$= (A+B+C)(A+B+\bar{C})(A+B+C)(\bar{A}+B+C)$$

$$(A+B+C)(A+\bar{B}+C)$$

$$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(A+\bar{B}+C)$$

Ans

Minterm & Maxterm

Minterm are defined as the product term which contain all the variable either in normal form or complement form.

Σm

$$\begin{matrix} A \rightarrow 1 \\ \bar{A} \rightarrow 0 \end{matrix}$$

Maxterm are defined as the sum term which contain all the variable either in normal form or complement form.

ΠM

$$\begin{matrix} A \rightarrow 0 \\ \bar{A} \rightarrow 1 \end{matrix}$$

Standard SOP

$$Y = AB\bar{C} + \bar{A}B\bar{C} + ABC$$

Minterm

POS

$$Y = (A+B+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$$

Maxterm

$$Y = \begin{matrix} \bar{A}\bar{B}\bar{C} & \bar{A}\bar{B}C & \bar{A}B\bar{C} & \bar{A}BC \\ 000 & 001 & 011 & 010 \\ m_0 & m_1 & m_3 & m_2 \end{matrix}$$

$$Y = \Sigma m(0,1,3,2)$$

$$Y = \begin{matrix} (A+B+\bar{C}) & (A+\bar{B}+\bar{C}) & (\bar{A}+\bar{B}+C) \\ 001 & 011 & 110 \\ M_4 & M_3 & M_2 \end{matrix}$$

$$Y = \Pi M(4,3,2)$$

$$Y = (A+B)(B+C)(C+A)$$

$$= (A+B+C\bar{C})(B+C+A\bar{A})(A+C+B\bar{B})$$

$$= (A+B+C)(A+B+\bar{C})(A+B+C)(\bar{A}+B+C)(A+\bar{B}+C)$$

$$= (A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(A+\bar{B}+C)$$

Minterm & Maxterm

Minterm are defined as the product term which contain all the variable either in normal form or complement form. Σm

$$\begin{matrix} A \rightarrow 1 \\ \bar{A} \rightarrow 0 \end{matrix}$$

Maxterm are define as the sum term which contain all the variable either in normal form or complement form. ΠM

$$\begin{matrix} A \rightarrow 0 \\ \bar{A} \rightarrow 1 \end{matrix}$$

Standard SOP

$$Y = AB\bar{C} + \bar{A}B\bar{C} + ABC$$

minterm

POS

$$Y = (A+B+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$$

maxterm

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C}$$

000

001

011

110

m0

m1

m3

m6

$$Y = \Sigma m(0,1,3,6)$$

$$Y = (A+B+\bar{C})(A+\bar{B}+\bar{C})(\bar{A}+\bar{B}+C)$$

001

011

110

m1

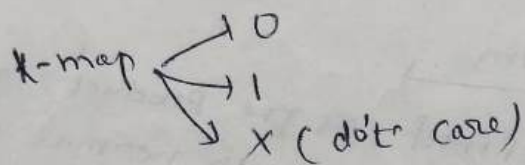
m3

m6

$$Y = \Pi M(1,3,6)$$

⑧ Karnaugh map (k-map) →

k-map is a graphical representation that provides a systematic method for simplifying & manipulating Boolean expression. This technique is the most extensively used tool for simplification of boolean fun.



Two variable →

$$n = 2$$

$$2^n = 2^2 = 4 \text{ cells required}$$

A \ B	0	1
0	00	01
1	10	11

Three variable

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

4-variable

AB \ CD	CD			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Fast solving K-map

- ① Octet
 - ② Quad
 - ③ Pairs
 - ④ Single Term
- Group

SOP form

1. $f(A,B) = \sum m(0,1,3)$

OR

$$Y = \bar{A}\bar{B} + \bar{A}B + AB$$

A \ B	0	1
0	1	1
1		1

$$Y = \bar{A} + B$$

2. $f(A,B) = AB + \bar{A}B + A\bar{B}$

A \ B	0	1
0		1
1	1	1

$$Y = B + A$$

$$f(A,B) = \sum m(0,3) + d(1)$$

$$d \begin{matrix} \nearrow 1 \\ \searrow 0 \end{matrix}$$

A \ B	0	1
0	1	1
1	0	1

$$Y = \bar{A} + B$$

$$C'Y(\bar{B}+B)$$

Q

$$f(ABC) = \sum m(1,2,3,5,6,7)$$

8

A \ BC	00	01	11	10
0	0	1	1	1
1	0	1	1	1

$$Y = C + B$$

Q

$$f(ABC) = \sum m(1,3,6,7)$$

A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

$$Y = \bar{A}C + AB$$

A \ BC	00	01	11	10
0	1	1	1	1
1	1	1	1	1

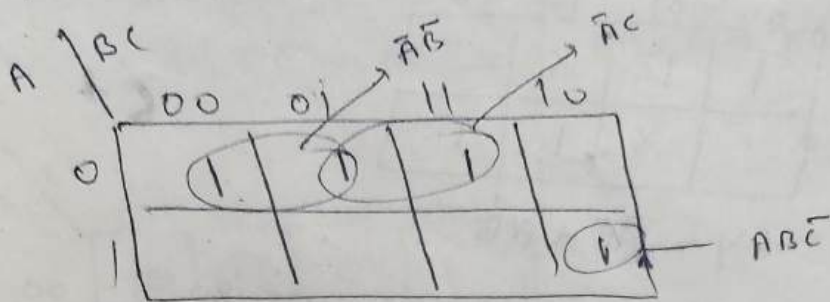
$$\bar{B} + \bar{A} + B\bar{C}$$

Q Q

$$Y = \bar{A} + A\bar{B} + AB\bar{C}$$

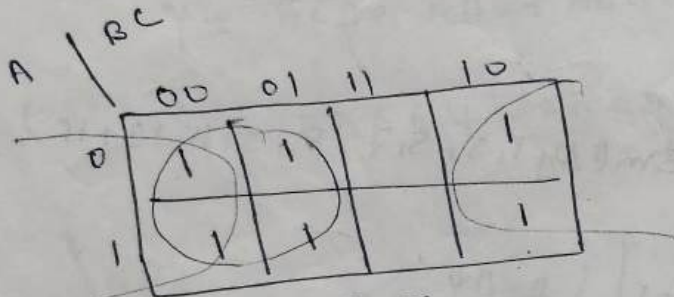
$$\begin{aligned} Y &= \bar{A}(B+\bar{B})(C+\bar{C}) + A\bar{B}(C+\bar{C}) + AB\bar{C} \\ &= (\bar{A}B + \bar{A}\bar{B})(C+\bar{C}) + A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} \\ &= \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} \end{aligned}$$

Q. $f(ABC) = \sum m(0, 1, 3, 6)$



$$Y = \bar{A}\bar{B} + \bar{A}C + AB\bar{C}$$

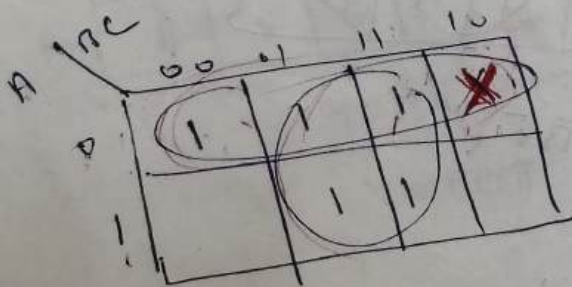
Q.



$$Y = \bar{B} + \bar{C}$$

Q.

$$f(ABC) = \sum m(0, 1, 3, 5, 7) + \sum d(2)$$

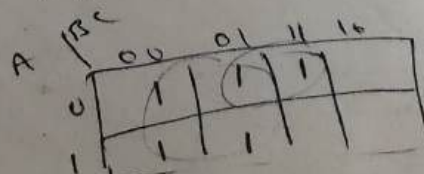


$$Y = \bar{A} + C$$

Q.

use a k-map to minimize the following standard sop expression

$$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}$$



$$\bar{B} + \bar{A}C$$

Q

		BC			
		00	01	11	10
A	0	1	1	X	
	1		X	1	1

$$Y = \bar{A}\bar{B} + AB$$

when do't care is not use so,
The value of do't care is 0.
If use do't care value of it 1.

Four Variable

$$f(ABCD) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$$

		CD			
		00	01	11	10
AB	00	1	1	1	0
	01	0	1	1	0
	11	0	1	1	0
	10	1	1	1	0

$$Y = A + \bar{B}\bar{C}$$

Q

$$Y = \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BCD + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + AB\bar{C}D + ABCD + A\bar{B}\bar{C}\bar{D}$$

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	1	1	1
11	1	1	1	0
10	0		1	0

Not necessary so redundant it

$$Y = \bar{A}\bar{C}D + \bar{A}BC + AB\bar{C} + ACD$$

Q

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

BD

$\bar{D}$

$$Y = B\bar{D} + \bar{B}\bar{D}$$

$$Y = B\bar{D} + \bar{B}\bar{D}$$

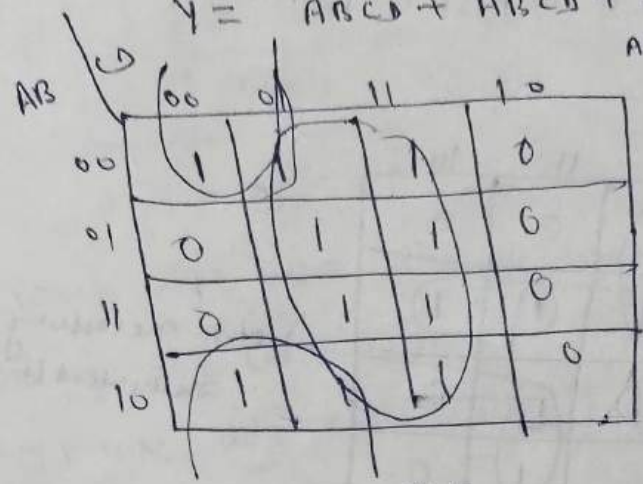
3185

0111 0111 1000 0011

Q

$f(ABCD) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$ OR

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD + ABC\bar{D} + ABCD$$



$$Y = D + \bar{B}C$$

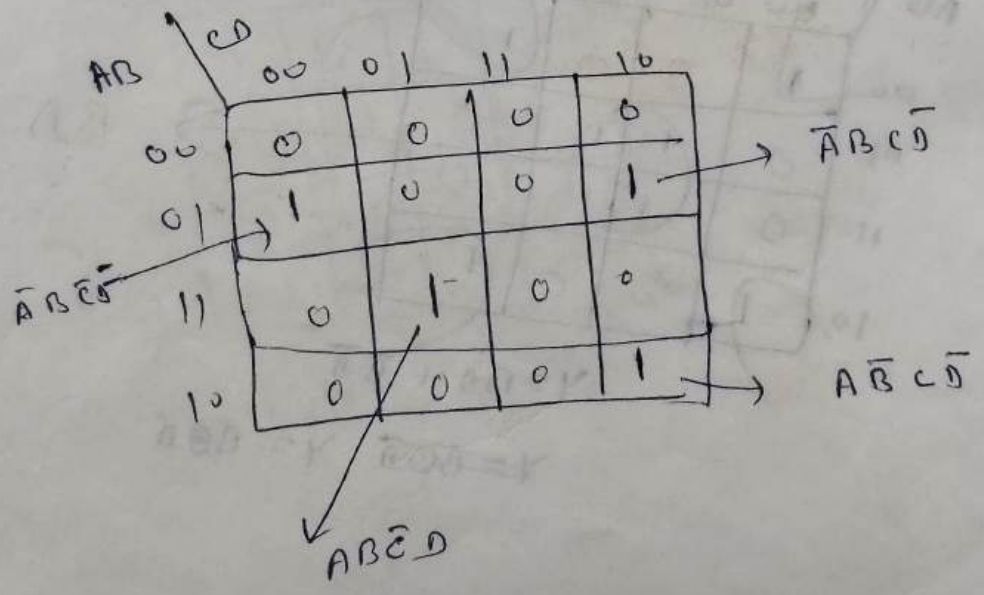
Q

$$Y = \bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D} + \bar{A}B\bar{C}D + A\bar{B}CD$$

Sol,

$$Y = 0100 + 1010 + 0110 + 1101$$

How fill



Q.

$$F(ABCD) = ACD + \bar{A}B + \bar{D}$$

first change in standard sop form

$$= ACD + \bar{A}B + \bar{D}$$

$$= \frac{ACD(B+\bar{B})}{\downarrow} + \frac{\bar{A}B(C+\bar{C})(D+\bar{D})}{\downarrow} + \frac{\bar{D}(B+\bar{B})(A+\bar{A})}{(C+\bar{C})}$$

$$= ABCD + A\bar{B}CD + (\bar{A}BC + \bar{A}B\bar{C})(D+\bar{D}) + \bar{D}$$

$$= ABCD + A\bar{B}CD + \bar{A}BCD + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D}$$

$$= ABCD + A\bar{B}CD + \bar{A}BCD + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D}$$

$$= ABCD + A\bar{B}CD + \bar{A}BCD + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D}$$

$$F = m(0, 2, 4, 5, 6, 7, 8, 10, 11, 12, 14, 15)$$

AB \ CD	00	01	11	10
00	1			1
01	1	1		1
11	1		1	1
10	1			1

$$F = \bar{D} + \bar{A}B + AC$$

Q $F(ABC) = \sum m(4, 5, 6, 10, 11, 12) + \sum d(2, 3, 7)$

AB \ CD	00	01	11	10
00	0	0	X	X
01	1	1	X	1
11	1	1	1	1
10	1	1	1	1

$\overline{A} \overline{B} \overline{C} D$

Q $Y(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$

AB \ CD	00	01	11	10
00	X	1	1	X
01	0	X	1	0
11	0	0	1	0
10	0	0	1	0

$\overline{A} \overline{B} + C D$
 $C D + \overline{A} D$

Q $Y(ABC) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$

				1
	1	X	1	1
		1		
X				

$\overline{B} D + \overline{C} D + A \overline{B} \overline{C} + A D$

$$\begin{aligned}
 Q \quad Y(A, B, C) &= AB + AB\bar{C} + \bar{B} \\
 &= AB(C + \bar{C}) + AB\bar{C} + \bar{B}(A + \bar{A})(C + \bar{C}) \\
 &= AB(C + AB\bar{C}) + AB\bar{C} + \bar{B}(A + \bar{A})(C + \bar{C}) \\
 &= ABC + AB\bar{C} + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} \\
 &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} \quad \text{--- (1)}
 \end{aligned}$$

Total Minterm are six

$$\begin{aligned}
 ABC(111) &= m_7 & AB\bar{C}(110) &= m_6, & A\bar{B}C(101) &= m_5 \\
 A\bar{B}\bar{C}(100) &= m_4 & \bar{A}\bar{B}C(001) &= m_1 & \bar{A}\bar{B}\bar{C}(000) &= m_0
 \end{aligned}$$

① can be written as sum of minterms

$$Y(A, B, C) = \sum m(0, 1, 4, 5, 6, 7)$$

POS in standard form

4

$$\begin{aligned}
 Y(A, B, C) &= (A + \bar{B})(A + \bar{B} + \bar{C}) \\
 &= (A + \bar{B} + C\bar{C})(A + \bar{B} + \bar{C}) \\
 &= (A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + \bar{B} + \bar{C})
 \end{aligned}$$

total maxterm are three

$$\begin{aligned}
 (A + \bar{B} + C)(010) &= m_2 \\
 (A + \bar{B} + \bar{C})(011) &= m_3 \\
 (A + B + \bar{C})(001) &= m_1
 \end{aligned}$$

$$Y(A, B, C) = \sum M(1, 2, 3)$$

base

2. Binary number 54(2) 01

3. octal number 54. (8) 0 1 1 2 3 4 5 6 7

4. Hexadecimal n.s (16) 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
10 11 12 13 14 15

Base, (Radix) It is defined as the no. of different digits which can occur in each position in the n.s.

Decimal number sy. has base of 10. It contains ten unique symbols (or digits). 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

$1 \mid 101.011$

↑ LSD

MSD least significant digit

most significant
digit

Binary to decimal conversion

9 → $(1101.011)_2 = ()_{10}$

$$1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1$$
$$1x^3 + 1x^2 + 0x^1 + 1x^0 \cdot 0x2^{-1} + 1x2^{-2} + 1x2^{-3}$$
$$8 + 4 + 0 + 1 \cdot 0 + \frac{1}{4} + \frac{1}{8}$$

$$(13.374)_3$$

convert $(1101)_2 = (\quad)_{10}$

1 1 0 0 1
16 8 4 2 1 $= (25)_{10}$

Decimal $\rightarrow$ Binary

$$(25)_{10} = ()_2$$

2	25	Remainders
2	12	1
2	6	0
2	3	0
2	1	1

$$(11001)_2$$

$$(0.8125)_{10} \rightarrow ()_2$$

$$0.8125 \times 2 = 1.625 = 0.625$$

$$0.625 \times 2 = 1.25 = 0.25$$

$$0.25 \times 2 = 0.50 = 0.50$$

$$0.50 \times 2 = 1.0 = 0.0$$

$$(0.8125)_{10} \rightarrow (0.1101)_2$$

Q. $(0.77)_{10} = ()_2$

$$0.77 \times 2 = 1.54 = 0.54 \quad \text{With a carry } 1$$

$$0.54 \times 2 = 1.08 = 0.08 \quad \text{''}$$

$$0.08 \times 2 = 0.16 = 0.16 \quad \text{''}$$

$$0.16 \times 2 = 0.32 = 0.32 \quad \text{''}$$

$$0.32 \times 2 = 0.64 = 0.64 \quad \text{''}$$

$$0.64 \times 2 = 1.28 = 0.28 \quad \text{''}$$

$$(0.77)_{10} = (0.110001)_2$$

Q. convert $(25.625)_{10} = ()_2$

2	25
---	----

$$(11001.101)_2$$

$$0.625 \times 2 = 1.25 = 0.25 \quad 1$$

$$0.25 \times 2 = 0.5 = 0.5 \quad 0$$

$$0.5 \times 2 = 1.0 = 0.0 \quad 1$$

Octal Number

base = 8

0, 1, 2, 3, 4, 5, 6, 7

$\leftarrow 8^3 \quad 8^2 \quad 8^1 \quad 8^0 \quad \cdot \quad 8^{-1} \quad 8^{-2} \quad 8^{-3} \rightarrow$
 ↑
 octal point

9. $(352)_8 = (\quad)_{10}$

$$3 \times 8^2 + 5 \times 8^1 + 2 \times 8^0$$

$$192 + 40 + 2 = (234)_{10}$$

Q. ~~(352)~~ $(27.24)_8 = (\quad)_{10}$

$$1 \times 8^2 + 2 \times 8^1 + 7 \times 8^0 + 2 \times 8^{-1} + 4 \times 8^{-2}$$

$$1 \times 64 + 2 \times 8 + 7 \times 1 + \frac{2}{8} + \frac{4}{64}$$

$$64 + 16 + 7 + \frac{2}{8} + \frac{4}{64}$$

$$(87.3125)_{10}$$

Decimal $\rightarrow$ octal conversion

$(175)_{10} = (\quad)_8$

8	175
8	21
	5

with 7 remainder

$(257)_8$

Q. $(0.15)_{10} = (\quad)_8$

$0.15 \times 8 = 1.20 = 0.20$ with a carry of 1

$0.20 \times 8 = 1.60 = 0.60$

$0.60 \times 8 = 4.80 = 0.80$

$(.114)_8$

Binary to octal conversion

Simplest procedure, the given binary no. is arranged into groups of 3 bits starting from the octal point & then each group is converted to its equivalent octal no. where necessary, extra 0s can be added in front (i.e. left end) of binary no. to complete group of three.

Q. $(101011)_2 = (\quad)_8$

$$\begin{array}{ccc} 101 & 011 & \\ \downarrow & \downarrow & \\ 5 & 3 & \end{array} = (53)_8$$

Q. $(111110111)_2 = (\quad)_8$

$$\begin{array}{ccc} 111 & 110 & 111 \\ \hline 7 & 6 & 7 \end{array} \quad (767)_8$$

Q. $(10101.11)_2$

add extra 0

$$\begin{array}{ccc} 010 & 101 & .110 \\ \hline & & \end{array} = (25.6)_8$$

The chief advantage of this system is that its no. are one-third in length as compared to binary no.

octal to binary conversion

$(75)_8 = (\quad)_2$

$(111101)_2$

$(74.562)_8$

$$\begin{array}{ccccccc} 111 & 100 & . & 101 & 110 & 010 & \end{array}$$

$(111100.101110010)_2$

Hexadecimal Number System

Base = 16

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
										10	11	12	13	14	15

Binary to Hexadecimal

The simple method is to split the given binary no. into 4-bit groups.

$$\begin{array}{ccccccc} (10001100)_2 & = & (&)_{16} \\ \downarrow & & \downarrow & & & & \\ 8 & & 12 & & & & (8C)_{16} \end{array}$$

$$(1011010111)_2$$

$$\begin{array}{ccc} 0010 & 1101 & 0111 \\ \downarrow & \downarrow & \downarrow \\ 2 & 13 & 7 \end{array}$$

$$(2D7)_{16}$$

Hexadecimal $\rightarrow$ Binary

$$(23A)_{16} = (?)_2$$

$$\downarrow$$
$$\underline{0010} \quad \underline{0011} \quad \underline{1010}$$

$$(23A)_{16} = (001000111010)_2$$

$$(23A)_{16} = (1000111010)_2$$

Addition	Carry	Sum
0+0 =	0	0
0+1	0	1
1+0	0	1
1+1	1	0

$$\begin{array}{r} \text{Q} \quad (101) \\ + (110)_2 \\ \hline 101 \\ + 110 \\ \hline 1011 \text{ Ans} \end{array}$$

Add $(110011)_2$ to $(101101)_2$

$$\begin{array}{r} 11111 \\ 110011 \\ + 101101 \\ \hline 1100000 \text{ Ans} \end{array}$$

Subs Subtraction

	borrow	
0-0	0	0
0-1	1	0
1-0	0	1
1-1	0	0

$$\begin{array}{r} \text{Q} \quad (1110)_2 - (0101)_2 \\ \begin{array}{r} 1110 \\ - 0101 \\ \hline 1001 \end{array} \end{array}$$

$$\begin{array}{r} \text{Q} \quad (1001) \\ - (011) \\ \hline (0010) \end{array}$$

1st column $1-1=0$
 2nd $0-1=1$ with 1 borrow
 3 $1(\text{borrow})-1=0$
 $0(\text{after borrow})-0=0$

Complem

Q. 10101×101

$$\begin{array}{r}
 10101 \\
 \times 101 \\
 \hline
 10101 \\
 00000 \\
 10101 \\
 \hline
 1010101
 \end{array}$$

$$\begin{array}{r}
 10101 \\
 \times 101 \\
 \hline
 10101 \\
 00000 \\
 10101 \\
 \hline
 1101001
 \end{array}$$

Q. 1101×101

$$\begin{array}{r}
 1101 \\
 \times 101 \\
 \hline
 1101 \\
 0000 \\
 1101 \\
 \hline
 100001
 \end{array}$$

Q. $101010 \div 100$

$$\begin{array}{r}
 100 \overline{) 101010} \\
 \underline{100} \\
 00101 \\
 \underline{100} \\
 100 \\
 \underline{100} \\
 000
 \end{array}$$

$110011 \div 11$

$$\begin{array}{r}
 11 \overline{) 110011} \\
 \underline{11} \\
 000011 \\
 \underline{11} \\
 00
 \end{array}$$

Complements →

Complements are often used in digital computer to simplify subtraction ^{& for logical manipulations.}. The complements are classified in two types on the number system base.

1. n 's complement

n = radix or base

$(n-1)$'s , ,

Decimal N . 10's complement
 9's , ,

Binary N . 2's
 1's

And 10's complement of decimal N .

Q. $(7634)_{10}$

1st And 9's complement

$$\begin{array}{r} 9999 \\ - 7634 \\ \hline 2365 \\ 10's \quad + 1 \\ \hline 2366 \end{array}$$

Q. $(76.34)_{10}$

1st And 9's complement

$$\begin{array}{r} 9999 \\ - 76.34 \\ \hline 23.65 \\ 10's \quad 23.66 \end{array}$$

find 2's complement for given binary no.

Q (11011)₂

Q (1101.01)₂

$$\begin{array}{r} \text{1's} \quad 11111 \\ - 11011 \\ \hline \text{1's} \quad 00100 \\ + \quad 1 \\ \hline \text{2's} \quad (00101)_2 \end{array}$$

2's 0010.11

finding the 1's complement

The 1's complement of a binary no. is found by changing all 1s to 0's & all 0's to 1s.

$$\text{2's complement} = \text{1's complement} + 1$$

Q Find 2's complement

10110010

$$\begin{array}{r} \text{1's} \quad 01001101 \\ + \quad 1 \\ \hline \text{2's} \quad 01001110 \end{array}$$

Subtraction with (r-1)'s complement

For subtraction of minuend & subtrahend (M-S),
both the no. with base r.

1. Equate the no. of digits in minuend & subtrahend
2. Find (r-1)'s comp to the subtrahend.
3. Add minuend with (r-1)'s comp. of subtrahend
→ discard the carry.
4. If the above addition produce a carry, add this carry to the LMS (most significant bit) of the sum.
5. If the above addition does not produce a carry, take (r-1)'s comp to the result of above sum & place negative sign in front. Ans. is in 2's comp form.

Perform sub. by using 9's comp.

$$76532 - 4250$$

$$\begin{array}{r} 76532 \\ - 04250 \\ \hline \end{array}$$

$$\begin{array}{r} 99999 \\ - 04250 \\ \hline 95749 \end{array}$$

$$\begin{array}{r} 76532 \\ + 95749 \\ \hline 172281 \\ \hline \end{array}$$

$$72282 \text{ Ans}$$

$$\begin{array}{r} 7 \\ -5 \\ +2 \\ \hline \end{array} \begin{array}{l} 0111 \\ -0101 \\ +1011 \\ \hline 10010 \end{array} \begin{array}{l} \text{is } 1010 \\ 2's \text{ } 0001 \\ 1011 \end{array}$$

0010 Ans.

$$\begin{array}{r} 5 \\ -7 \\ -2 \end{array}$$

$$\begin{array}{r} 0101 \\ -11 \\ \hline 001 \end{array}$$

$$\begin{array}{r} 101 \\ +001 \\ \hline 110 \end{array}$$

No. carry mean
take again 2's

$$001$$

Q

4250 - 76532

$$\begin{array}{r} 04256 \\ - 76532 \\ \hline \end{array}$$

9's comp of 76532

$$\begin{array}{r} 99999 \\ - 76532 \\ \hline 23467 \end{array}$$

9's comp

$$\begin{array}{r} 04256 \\ + 23467 \\ \hline 27717 \end{array}$$

No carry, again take 9's comp. of the result

$$\begin{array}{r} 99999 \\ 27717 \\ \hline - 72282 \end{array} \quad \underline{\underline{\text{Ans}}}$$

Q.

11011 - 1010

$$\begin{array}{r} 11011 \\ - 01010 \rightarrow 1's \\ \hline \end{array}$$

$$\begin{array}{r} 11011 \\ + 10000 \\ \hline 110101 \end{array}$$

dis

$$\begin{array}{r} 11011 \\ + 10101 \\ \hline 110000 \\ \downarrow + 1 \\ 10001 \end{array}$$

10001 Ans

Q

1010 - 1101

$$\begin{array}{r} 01010 \\ - 11011 \rightarrow 1's \\ \hline \end{array}$$

$$\begin{array}{r} 01010 \\ + 00100 \\ \hline 01110 \end{array}$$

No carry, take again 1's

$$- 10001 \quad \underline{\underline{\text{Ans}}}$$

Complements of Numbers

Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.

Simplifying operation leads to simpler, less expensive ckt to implement the operations.

There are two types of complements for each base r system.

r 's complement (radix complement)

$(r-1)$'s complement. (The diminished radix comp.)

Binary $\begin{cases} \rightarrow 1's \\ \rightarrow 2's \end{cases}$

Octal $\begin{cases} \rightarrow 7's \\ \rightarrow 8's \end{cases}$

Decimal $\begin{cases} \rightarrow 9's \\ \rightarrow 10's \end{cases}$

Hexa $\begin{cases} \rightarrow 15's (F) \\ \rightarrow 16's \end{cases}$

→ To determine $(r-1)$'s complement subtract given no. from $(r^n - 1)$. where r is the base & n is the no. of bits.

Ex. 1's complement of 10110

first write highest no. of binary

i.e.

$$\begin{array}{r} 11111 \\ 10110 \\ \hline 01001 \end{array}$$

Then subtract the given no.

Ex. 7's complement of octal 245

777 → highest no. in octal

$$\begin{array}{r} 777 \\ - 245 \\ \hline 532 \end{array}$$

Ex. 9's complement of decimal 2453

9999 → highest no. in decimal

$$\begin{array}{r} 9999 \\ - 2453 \\ \hline 7546 \end{array}$$

Signed

Ex. F's complement of 3679

$$\begin{array}{r} \text{FFFF} \\ - 3679 \\ \hline \text{C986} \end{array}$$

To determine 8's complement, first write 8-1's complement & then add 1 to the LSB (lower significant bit)

Ex. 2's comp $\rightarrow$ 10110

$$\begin{array}{r} 1's \rightarrow 01001 \\ + 1 \\ \hline 01010 \end{array}$$

Ex. 2's complement of binary 11010.11

$$1's \text{ comp} \rightarrow 00101.00$$

$$\begin{array}{r} 00101.01 \end{array}$$

* 1 is always add in rightmost

Exp 8's complement of 3470

7's compl. of 3470

$$\begin{array}{r} 7777 \\ - 3470 \\ \hline 4307 \\ + 1 \end{array}$$

$$8's \text{ comp. } 4310$$

Exp. 16's comp $\rightarrow$ 2369

$$\begin{array}{r} \text{FFFF} \\ - 2369 \\ \hline \text{DC96} \\ + 1 \end{array}$$

15's

$$16's \rightarrow \text{DC97}$$

Signed Binary Numbers

Positive integers (including zero) can be represented as unsigned numbers. However, to represent negative integers, we need a notation for negative values.

Signed - magnitude representation

Signed - 1's complement representation

Signed - 2's complement representation

Signed - magnitude rep.

In decimal no system a plus (+) sign is used to denote a positive no. and a minus (-) sign for denoting a negative number. The plus sign is usually dropped, and the absence of any sign means that n. has +ve value. This representation of no. is known as signed no.

Digital ckt can understand only two symbols, 0 & 1; therefore, we must use the same symbols to indicate the sign of the no. also.

Normally, an additional bit is used as the sign bit and it is placed as the most significant bit.

0 is used to represent a positive no. & 1 to ~~represent~~ represent a negative no.

It is important to realize that both signed and unsigned binary no. consists of a string of bits when represented in a computer. The user determine whether the no. is signed or unsigned. If binary n. is signed, then left most bit represent the sign & rest of bit represent the no.

For exp. The string of bits 01001 can be considered as +9 (unsigned binary) OR +9 (signed binary) because left most bit is 0.

The string of bits 11001 represents the binary equivalent of 25 when considered as an unsigned no. &

The binary equivalent of -9 when considered as a signed no. bcoz

$$\begin{array}{r} 11001 \\ \downarrow \quad \downarrow \\ -ve \quad -9 \end{array}$$

usually, there is no confusion in interpreting the bits if the type of representation for the no. is known in advance.

It is referred to as the signed-magnitude convention. This is the representation of signed no. used in ordinary arithmetic.

When arithmetic operations are implemented in a computer, it is more convenient to use a different system, ~~also~~ referred to as sign-magnitude system.

for representing a negative signed no.

In this sy. a negative no. is indicated by its complement.

Ex. 9, represent in binary with sign bit
+9 is represented with sign bit of 0 in the
leftmost position, 00001001.

-9

Signed-magnitude rep. 10001001

Signed-1's compl. 11110110

Signed-2's comp. 11110111

In signed-magnitude, -9 is obtained from
+9 by changing only the sign bit in the
leftmost position from 0 to 1.

In signed 1's comp., -9 is obtained by complementing
all the bits of +9, including the sign bit.

The signed 2's comp. rep. of -9 is obtained by
taking the 2's comp. of the +ve no., including
the sign bit.

Q8. one's complement

$(0101)_2$ represent $(+5)_{10}$ where $(1010)_2$ represent $(-5)_{10}$ in ~~this~~ one's comp. rep.

$(0101)_2 = (+5)_{10}$ Therefore 1011 rep $(-5)_{10}$ in 2's comp rep.

Q9. And the decimal equivalent of the following bi

101100

sign bit is 1, which means the no. is -ve

Mag. $01100 = (12)_{10}$

$(-12)_{10}$

001000

$(+8)_{10}$

Q10. find the one's comp. of

rep. the following no. in one's comp. form

(a) $+7$ or -7

$(+7)_{10} = (0111)_2$ & $(-7)_{10} = (1000)_2$

$(+15)_{10} = (01111)_2$ & $(-15)_{10} = (10000)_2$

Represent $(-17)_{10}$ in

- (a) sign-mag.
- (b) one's comp
- (c) two comp.

The minimum no. of bits required to rep
 $(+17)_{10}$ in signed no. ~~is~~ format is six

$$(+17)_{10} = (010001)_2$$

Then $(-17)_{10}$

- (i) 110001 in sign-mag. representation
- (ii) 101110 in 1's comp
- (iii) 101111 in 2's comp.

For an n -bit number, the max positive no. which can be represent in 2's comp form is $(2^{n-1}-1)$ and the max negative no. is -2^{n-1} . It's a 2's comp no. represented by 4-bit binary no. it is observed that the max positive no is $0111 = +7$, whereas the max negative no. is $1000 = -8$ using four bit in 2's comp. format.

Sign-magnitude, 1's & 2's complement representation using 4-bits.

Decimal no.	Sign-magnitude	one's comp.	Two's comp.
0	0000	0000	0000
1	0001	0001	0001
2	0010	0010	0010
3	0011	0011	0011
4	0100	0100	0100
5	0101	0101	0101
6	0110	0110	0110
7	0111	0111	0111
-8	1000	1000	1000
-7	1001	1001	1001
-6	1010	1010	1010
-5	1011	1011	1011
-4	1100	1100	1100
-3	1101	1101	1101
-2	1110	1110	1110
-1	1111	1111	1111
-0	0000	0000	0000

Q Represent the following no. in one's comp. form

③ $+7$ & -7 ④ $+8$ & -8

③ $(+7)_{10} = (0111)_2$ & $(-7)_{10} = (1000)_2$

④ $(+8)_{10} = (01000)_2$ & $(-8)_{10} = (10111)_2$

Q Represent $(-17)_{10}$ in sign-magnitude, one's & Two's comp

The minimum no. of bits required to represent $(-17)_{10}$ in signed no. format is six.

$$(+17)_{10} = (010001)_2$$

Therefore $(-17)_{10}$ is represented by

(i) 110001 in sign-magnitude repr.

(ii) 101110 in 1's comp. repr.

(iii) 101111 in 2's , , ,

$$X = 1010100$$

$$Y = 1000011$$

$X - Y$? $Y - X$? using 2's complements

$$\begin{array}{r} X - Y \\ 1010100 \\ - 1000011 \\ \hline \end{array}$$

2's comp of Y

$$\begin{array}{r} 0111100 \\ + 1 \\ \hline 0111101 \end{array}$$

$$\begin{array}{r} X = 1010100 \\ + 0111101 \\ \hline \text{Sum } 10010001 \end{array}$$

If carry present, discard carry

$$0010001 \text{ Ans}$$

$$\begin{array}{r} Y - X \\ 1000011 \\ - 1010100 \\ \hline \end{array}$$

2's comp of X

$$\begin{array}{r} 0101011 \\ + 1 \\ \hline 0101100 \end{array}$$

$$\begin{array}{r} 1000011 \\ + 0101100 \\ \hline 1101111 \end{array}$$

There is no end carry. Therefore, the ans is $Y - X = - (2's \text{ complement of } 1101111)$

$$- 0010001 \text{ Ans}$$

Q

$$\begin{array}{r} 1011 \\ - 1001 \\ \hline \end{array}$$

$$\begin{array}{r} 1001 \\ - 1011 \\ \hline \end{array}$$

2's comp. of (1001)

$$\begin{array}{r} \text{1's comp} \quad 0110 \\ + 1 \\ \hline \text{2's} \quad 0111 \end{array}$$

$$\begin{array}{r} 111 \\ 1011 \\ + 0111 \\ \hline 10010 \end{array}$$

discard carry 0010 Ans

$$\begin{array}{r} 1001 \\ - 1011 \\ \hline \end{array}$$

2's comp of (1011)

$$\begin{array}{r} \text{1's} \quad 0100 \\ + 1 \\ \hline \text{2's} \quad 0101 \end{array}$$

$$\begin{array}{r} 1 \\ 1001 \\ + 0101 \\ \hline 1110 \end{array}$$

~~There is carry~~ is no
There is no carry.

Ans - (2's comp of 1110)

$$\begin{array}{r} \text{1's} \quad 0001 \\ + 1 \\ \hline \text{2's} \quad 0010 \end{array}$$

Ans - 0010

72532 - 3250 using 10's comp.

$$M = 72532$$

$$N =$$

$$\begin{array}{r} 10's \text{ comp. of } 3250 \quad 9999 \\ - 3250 \\ \hline 6749 \\ 10's \quad \hline 6750 \end{array}$$

$$\begin{array}{r} M = 72532 \\ 10's \text{ comp. of } 96 \end{array}$$

$$m = 72532$$

$$N = 03250$$

$$10's \text{ comp. of } 03250$$

$$1'st \quad 9's$$

$$\begin{array}{r} 99999 \\ - 03250 \\ \hline 96749 \\ +1 \\ \hline 96750 \end{array}$$

$$10's$$

$$\begin{array}{r} M = 72532 \\ 10's \text{ comp. of } N = 96750 \\ \hline 169282 \end{array}$$

Sum

Discard end carry

$$\text{Ans } \underline{\underline{69282}}$$

$$3250 - 72532$$

$$M = 63250$$

$$N = 72532$$

$$10's \text{ comp of } N = 27468$$

$$M = 63250$$

$$10's \text{ comp of } N = 27468$$

$$\text{Sum} = 30718$$

There is no carry. Therefore, the ans is

$$-(10's \text{ comp. of } 30718) = -\underline{69282} \quad \underline{\text{Ans}}$$

(a) $X - Y = 1010100 - 1000011$ using 1's comp.

$$X = 1010100$$

$$1's \text{ comp of } Y = 0111100$$

$$\text{Sum} = 10010000$$

$$\text{End-around carry} + 1$$

$$X - Y = 0010001$$

(b) $Y - X$

$$Y = 1000011$$

$$1's \text{ comp of } X = 0101011$$

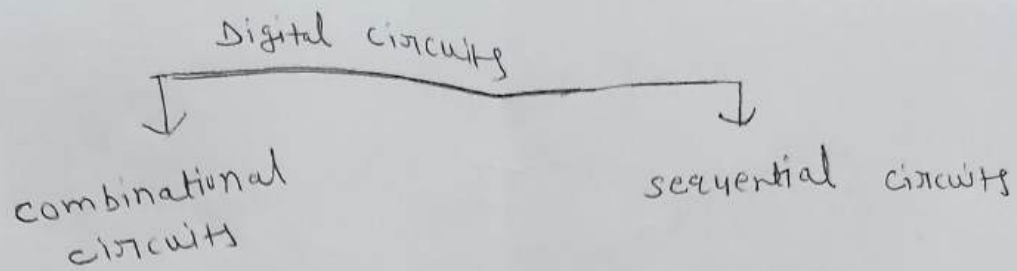
$$\text{Sum} = 1101110$$

There is no end carry. Therefore, the ans.

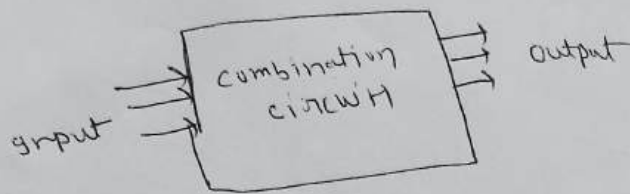
$$Y - X = -(1's \text{ comp of } 1101110)$$

$$= -\underline{0010001}$$

Ans



1. Present output depends on only present inputs.



2. No feedback is present
3. No memory in combinational circuit
4. No clock control is required in c.s.

5. Example

Half adder, full adder,
Half subtractor,
Full subtractor
Multiplexer etc.

1. Present output ~~depend~~ depends on present input and ~~it~~ also on previous output.



- (i) feedback is present
- (ii) Memory is present in sequential circuit
- (iv) clock control is present in S.C to synchronize

Example: (a) flip flop
(b) Register
(c) Counters etc

Combinational Logic circuits — output at any instant of time depend upon the input present at that instant of time.

Combinational Logic circuits do not have memory element. It can be designed using gates or available I.C.

Arithmetic ckt

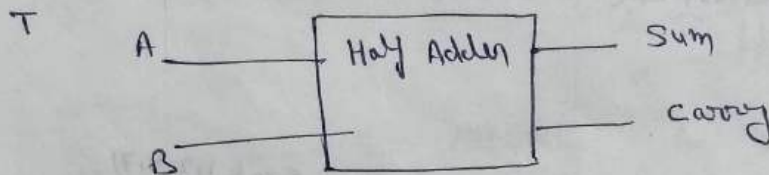
Half adder (HA)

H.A is a combination ckt which is used as two input & provide two o/p

i.e sum & carry.

Minimum no. of NAND = 5

NOR = 5



Truth table

A	B	Cy Carry	S Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Logical expression

$$S = \bar{A}B + A\bar{B} = A \oplus B$$

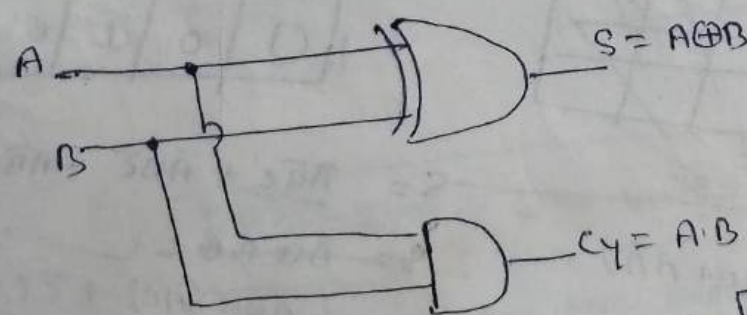
$$Cy = AB$$

$$S = \bar{A}B + A\bar{B}$$

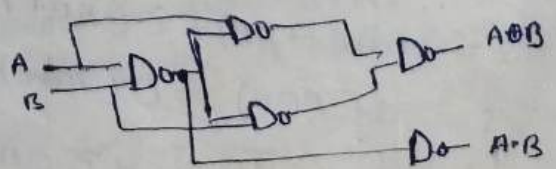
$$S = A \oplus B$$

$$Cy = A \cdot B$$

Implement



Half adder



c₄ =

Full adder (FA) →



FA is a combination ckt, which have 3 input & 2 output.

Truth Table →

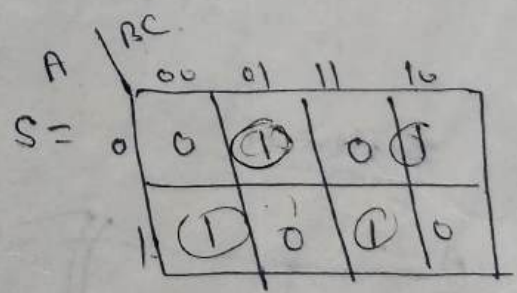
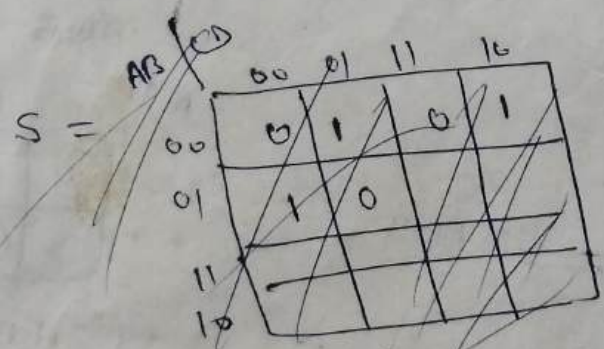
	A	B	C	c ₄	S
0 →	0	0	0	0	0
1 →	0	0	1	0	1
2 →	0	1	0	0	1
3 →	0	1	1	1	0
4 →	1	0	0	0	1
5 →	1	0	1	1	0
6 →	1	1	0	1	0
7 →	1	1	1	1	1

$$S = \sum m(1, 2, 4, 7)$$

$$c_4 = \sum m(3, 5, 6, 7)$$

$$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$C(\bar{A}\bar{B} + AB) + \bar{C}(\bar{A}B + A\bar{B})$$



$$S = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$S = C(\bar{A}\bar{B} + AB) + \bar{C}(\bar{A}B + A\bar{B})$$

$$S = C(A \odot B) + \bar{C}(A \oplus B)$$

$$S = C(\overline{A \oplus B}) + \bar{C}(A \oplus B)$$

$$S = C \oplus A \oplus B = A \oplus B \oplus C$$

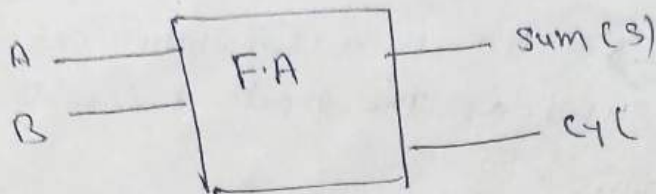
$$S = A \oplus B \oplus C$$

$$S = C(\bar{A}\bar{B} + AB) + \bar{C}(\bar{A}B + A\bar{B})$$

$$S = C(\bar{A}B + A\bar{B}) + \bar{C}(\bar{A}B + A\bar{B})$$

$$S =$$

Full adder (FA) →



FA is a combination ckt, which have 3 input & 2 output.

Truth Table →

	A	B	C	Cy	S
0 →	0	0	0	0	0
1 →	0	0	1	0	1
2 →	0	1	0	0	1
3 →	0	1	1	1	0
4 →	1	0	0	0	1
5 →	1	0	1	1	0
6 →	1	1	0	1	0
7 →	1	1	1	1	1

$$S = \sum m(1, 2, 4, 7)$$

$$C_y = \sum m(3, 5, 6, 7)$$

$$\begin{aligned} S &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\ C_y &= C(\bar{A}\bar{B} + AB) + \bar{C}(A\bar{B} + A\bar{B}) \end{aligned}$$

~~$S =$~~

AB	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

A \ BC	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$S = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

~~$$S = A \oplus B \oplus C$$~~

~~$$S = C(\bar{A}\bar{B} + AB) + \bar{C}(A\bar{B} + A\bar{B})$$~~

~~$$S = C(\bar{A}\bar{B} + AB) + \bar{C}(A\bar{B} + A\bar{B})$$~~

~~$$S =$$~~

$$S = C(\bar{A}\bar{B} + AB) + \bar{C}(A\bar{B} + A\bar{B})$$

$$S = C(A \oplus B) + \bar{C}(A \oplus B)$$

$$S = C(A \oplus B) + \bar{C}(A \oplus B)$$

$$S = C \oplus A \oplus B = A \oplus B \oplus C$$

$Cy =$

		BC			
		00	01	11	10
0		0	0	1	0
1		0	1	1	1

$$Cy = AB + BC + CA$$

$$Cy = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$AB + (\bar{A}B + A\bar{B})C$$

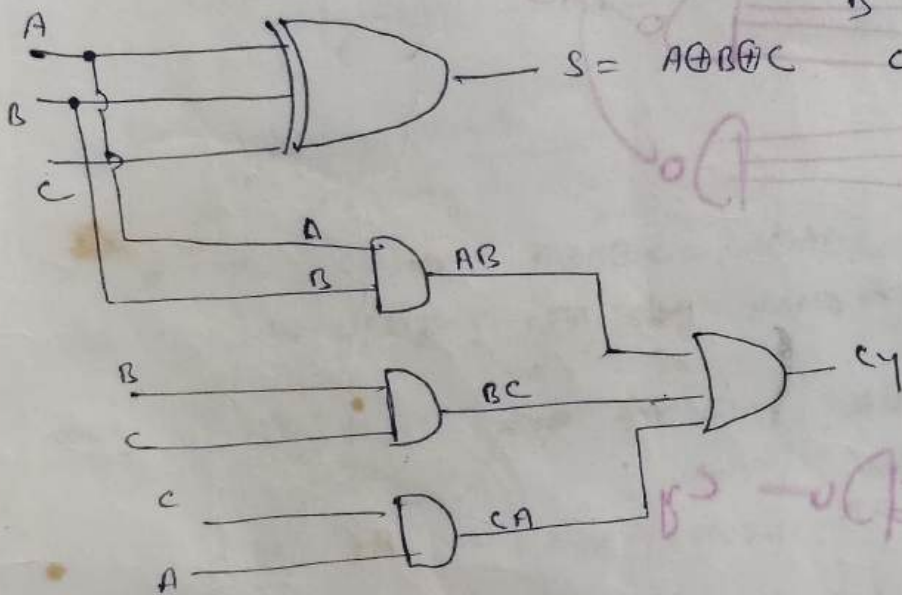
$$(AB + (A \oplus B)C)$$

it is for FA using HA

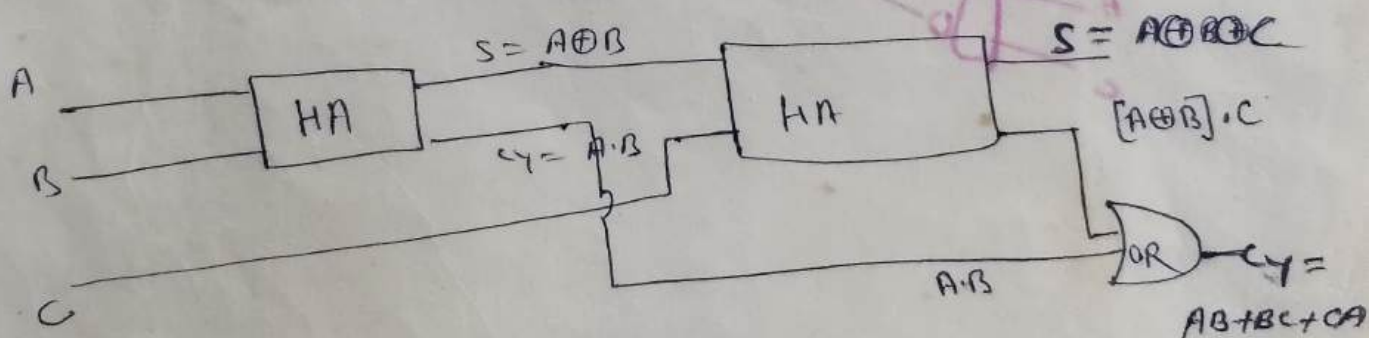
Total = 2 tpd

$$S = A \oplus B \oplus C$$

$$Cy = AB + BC + CA$$



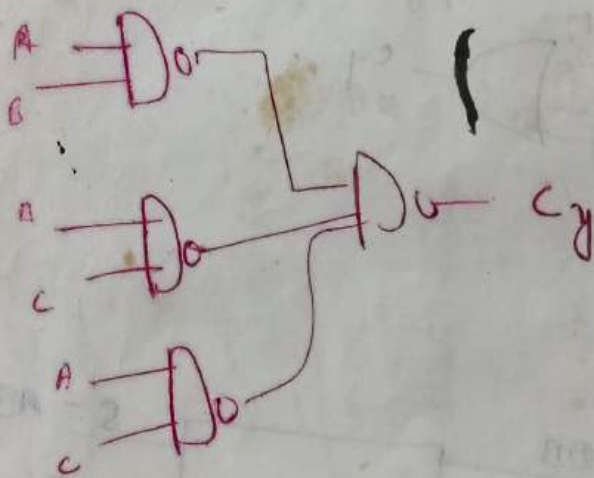
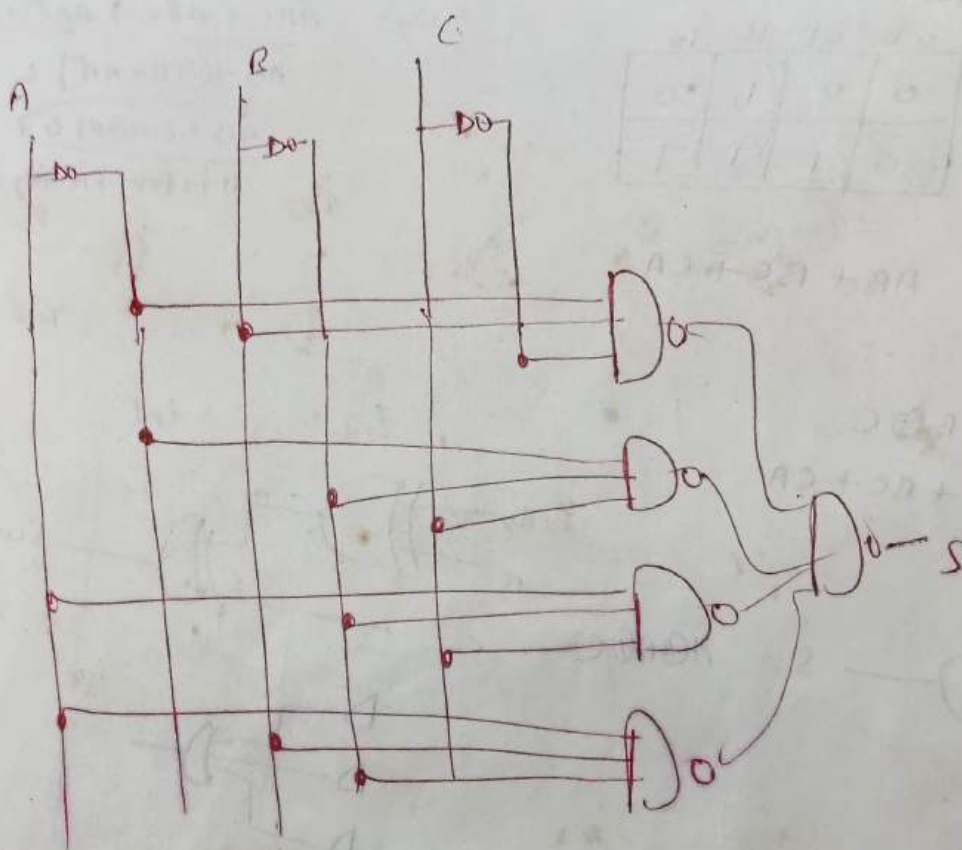
HA → FA



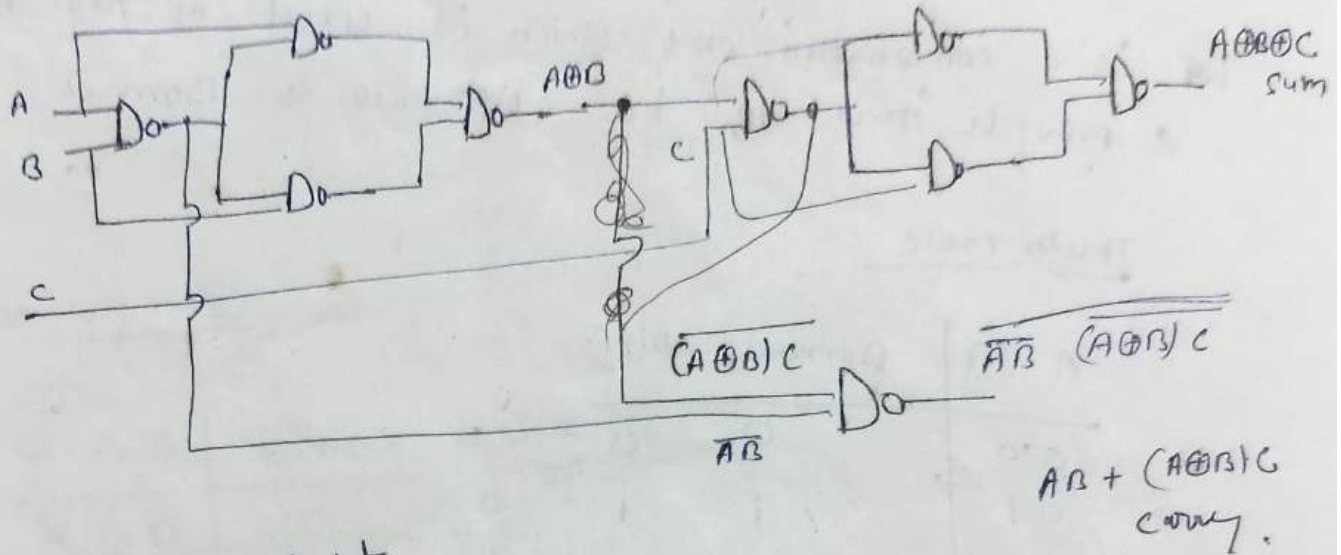
$$FA = 2HA + 1OR \text{ gate}$$

→ In FA each logic gate have propagation delay (tpd)
Then to provide sum or, carry out, it requires 2-logic
gate delay.

→ This CKT is for minimum pro-delay.



Full adder using NAND gate:-



Same as using NOR gate

* In 3-variable $A \oplus B \oplus C = A \oplus B \oplus C$
 * for designing FA using NAND or NOR is same.

- NOTE
- ① logic exp for sum $A \oplus B \oplus C$ or $A \oplus B \oplus C$
 carry AB
 $AB + BC + AC$ or $AB + (A \oplus B)C$
 - ② FA use 2 HA + 1 OR gate.
 - ③ Min no. of NAND gate = 9
 NOR = 9.

Half Subtractor (HS).

HS is a combination ckt which is used as two I/P & provide two OP i.e. Difference & Borrow

Truth table

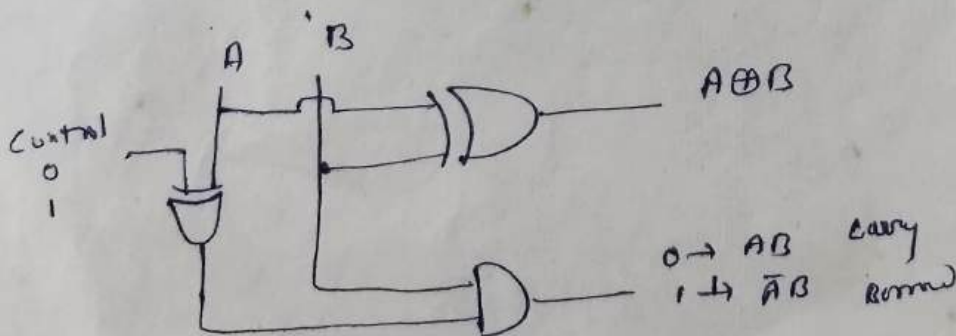
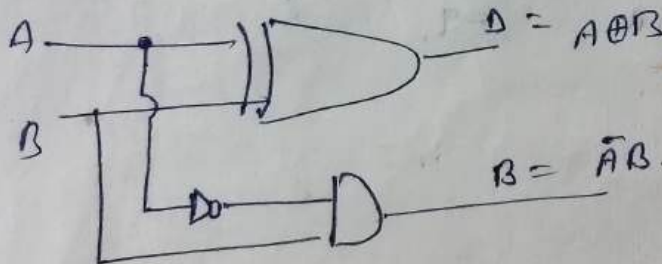
A	B	Borrow	Diff.
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

logical expression

$$\text{Difference} = \bar{A}B + A\bar{B}$$

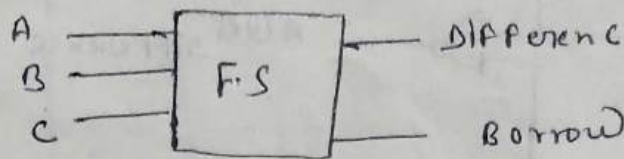
$$\text{Borrow} = \bar{A}B$$

logic ckt



This ckt can use as both H.A & H.S
 when control is 0 → ckt is HA
 1 → HS

Full Subtractor



Truth table

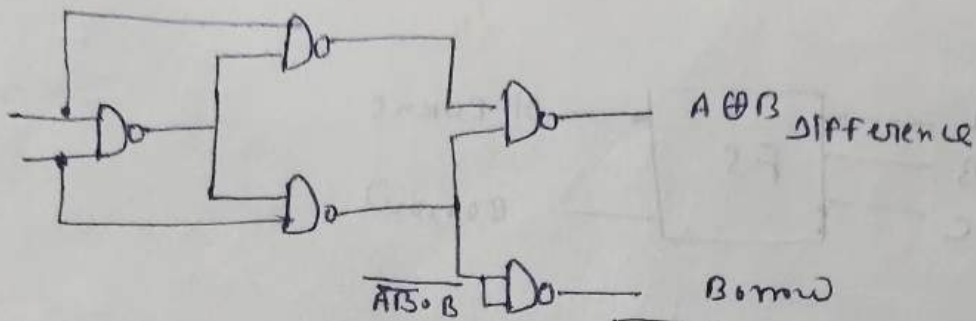
	A	B	C	Borrow	Difference
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	1	0
4	1	0	0	0	1
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$$\begin{aligned}
 \text{Diff} &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\
 &= A \oplus B \oplus C = \Sigma m(1, 2, 4, 7)
 \end{aligned}$$

$$\begin{aligned}
 \text{Borrow} &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC \\
 &= \bar{A}C + \bar{A}B + BC \\
 &= \bar{A}B + \bar{A}C + BC = \Sigma m(1, 2, 3, 7)
 \end{aligned}$$

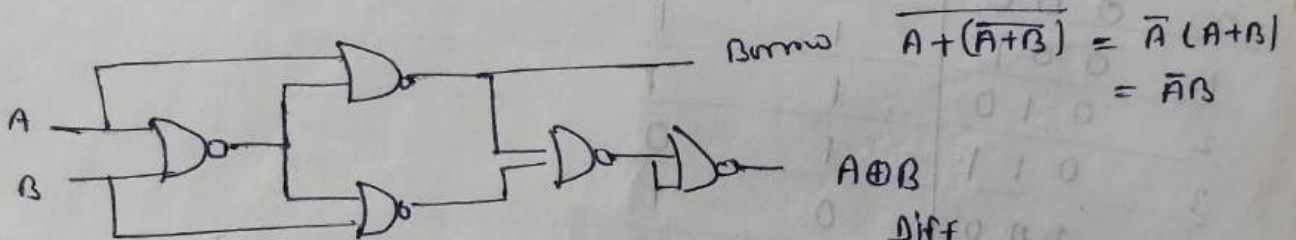
$$\begin{aligned}
 \text{also Borrow} &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + ABC \\
 &= \bar{A}\bar{B}C + \bar{A}B(\bar{C} + C) + ABC \\
 &= \bar{A}B + (A \oplus B)C
 \end{aligned}$$

Half subtractor using NAND gate -



$$\begin{aligned} \overline{A \cdot B} &= (\overline{A} + \overline{B}) \cdot B \\ &= \overline{A} B \end{aligned}$$

HS. using NOR gate



$$\begin{aligned} \overline{A + (\overline{A} + \overline{B})} &= \overline{A} (A + B) \\ &= \overline{A} B \end{aligned}$$

$A \oplus B$

Difference

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

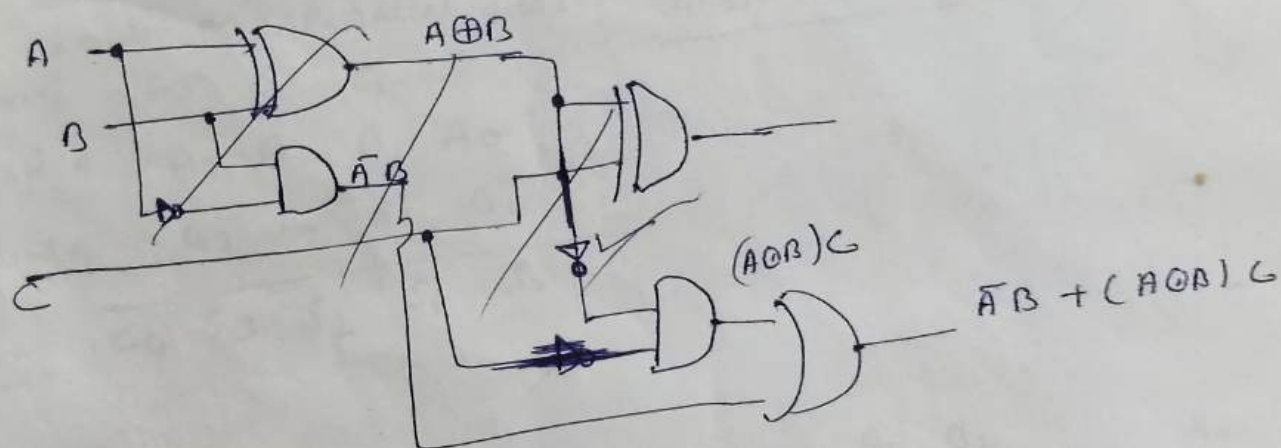
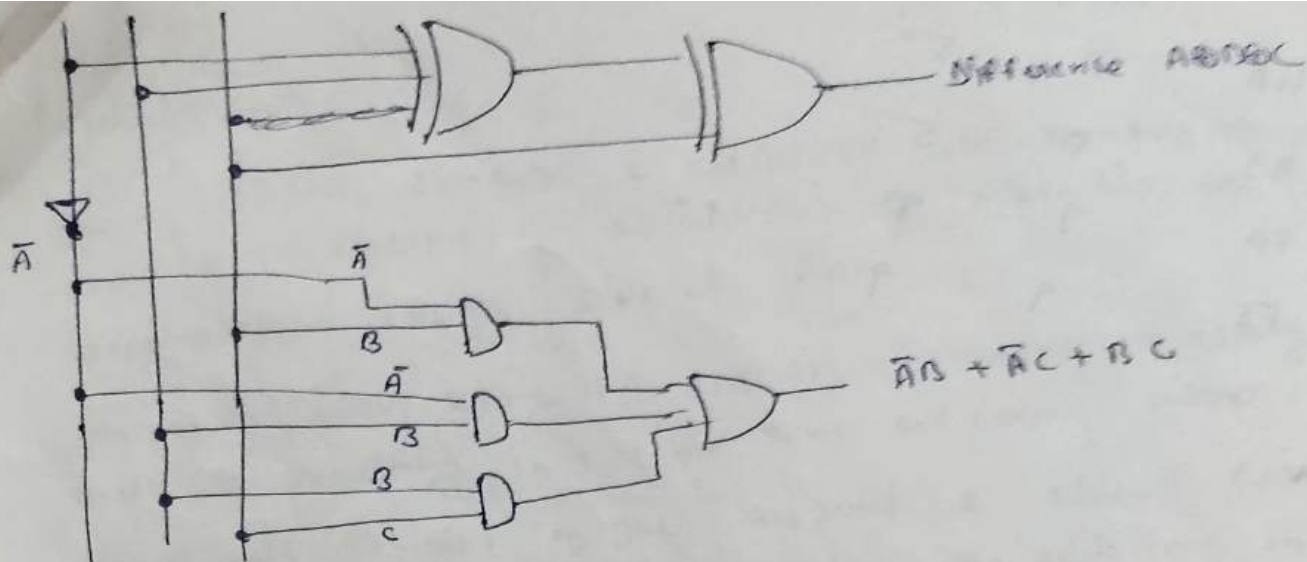
$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$

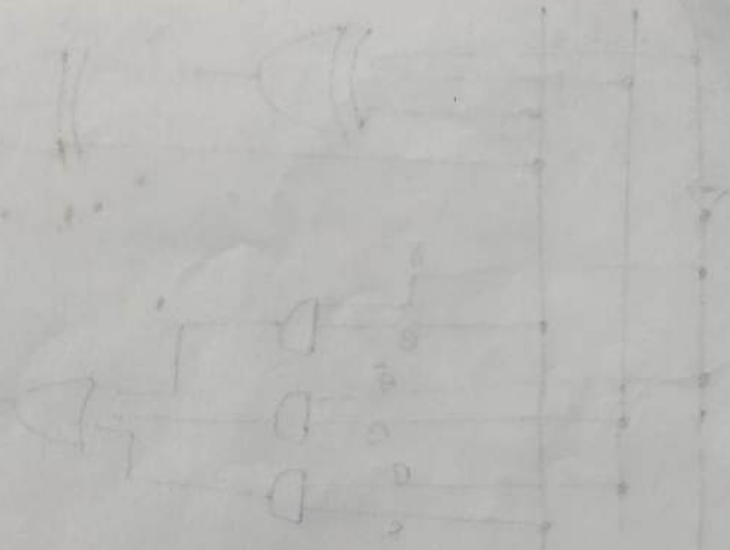
$$\sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4) = \sum m(1, 2, 3, 4)$$



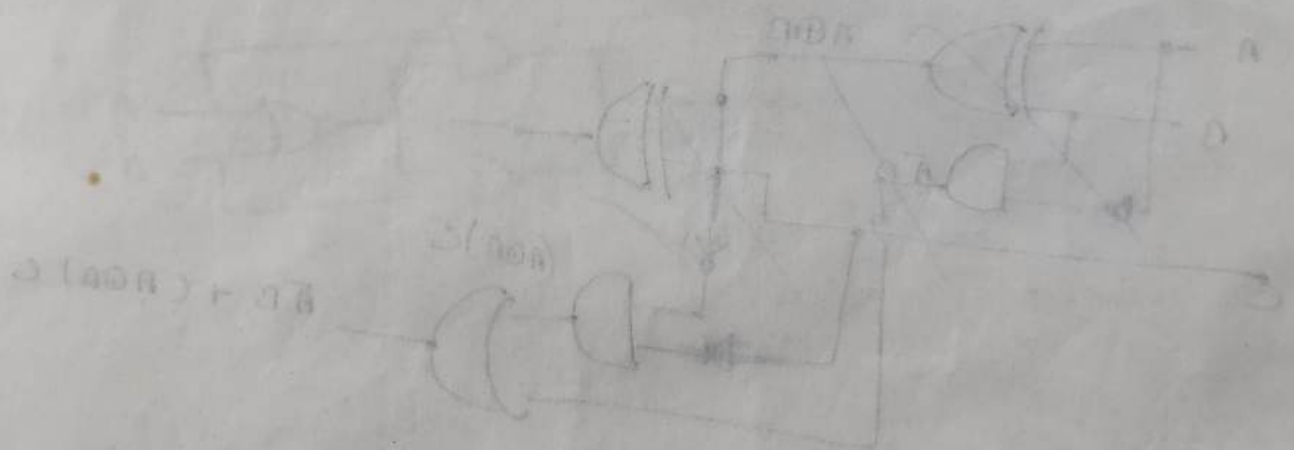
$$FS = 2HS + 1OR$$

	NAND	NOR
HA	5	5
HS	5	5
FA	9	9
FS	9	9

$$F_1 = AB + \bar{A}B + A\bar{B}$$



Parallel Adder



$$F_2 = AB + \bar{A}\bar{B}C$$

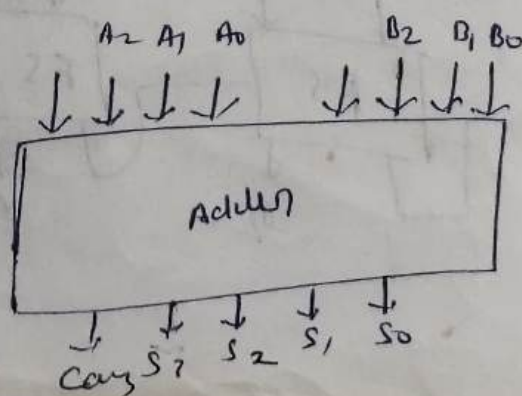
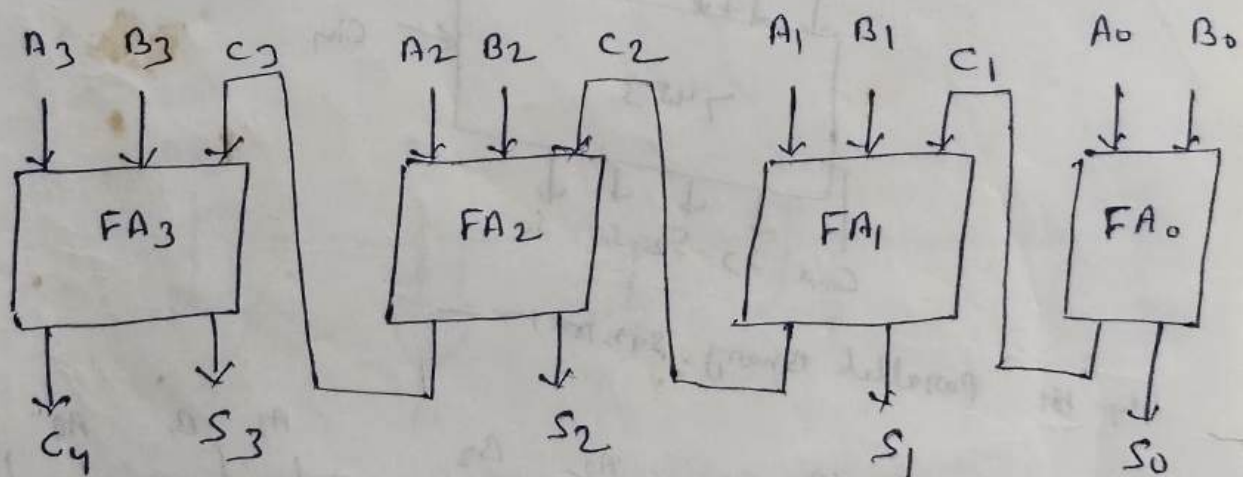
Parallel Adder ckt

~~Part~~ Modern computer & calculator use numbers ranging from 8 to 64 bits. The addition of multi bit no. can be accomplished using several F.A.

The 4-bit adder using FA circuit is capable of adding two 4-bit no. resulting in a 4-bit sum and carry output.

Since all the bits of the augend & addend are fed into the adder ckt simultaneously & the additions in each position are taking place at the same time, this ckt is known as parallel adder. 4-bit parallel adder →

Carry	C_3	C_2	C_1		
Augend A	A_3	A_2	A_1	A_0	
Addend B	B_3	B_2	B_1	B_0	
Sum	C_4	S_3	S_2	S_1	S_0



* Parallel adder is used to add group of bits.

To add two (n -bit) no., it requires,

$(n-1)$ FA, & 1 HA

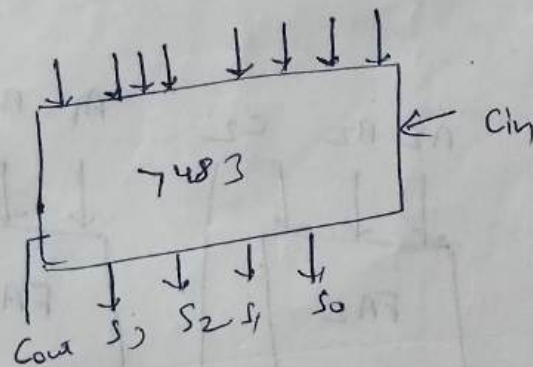
OR n FA

OR $(2n-1)$ FA & $(n-1)$ OR

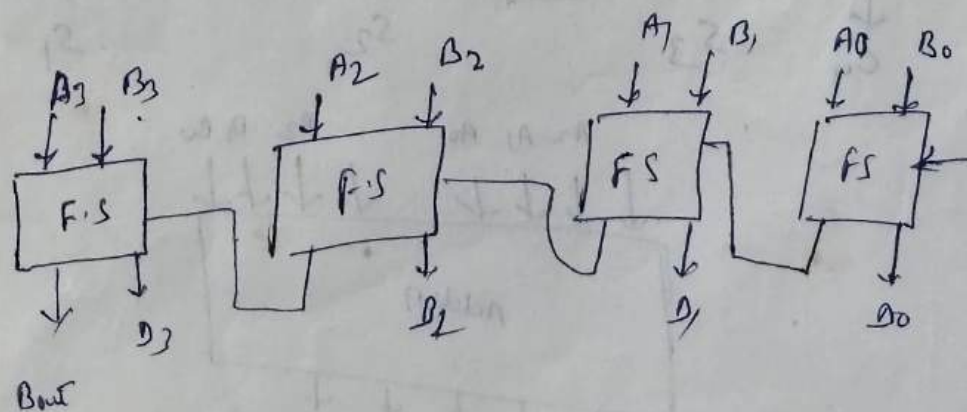
In parallel adder each FA provide 2 logic gate delays

In n -bit parallel adder $2n$ logic gate delays, require to provide result.

Disadvantage of parallel adder is carry propagation delay will present from i/p to o/p. due to this speed of operation will decrease as no. of bits are increased.



4-bit Parallel Binary Subtractor —



1-bit Parallel Adder/Subtractor

Perform the operations of both addition & subtraction.

Controlled inverter

The subtraction of two binary no. may be accomplished by taking 2's comp of the subtrahend & adding to the minuend. by this procedure, the subtraction becomes an addition operation. The 2's comp of the subtrahend can be obtained by adding 1 to the 1's comp of the subtrahend.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



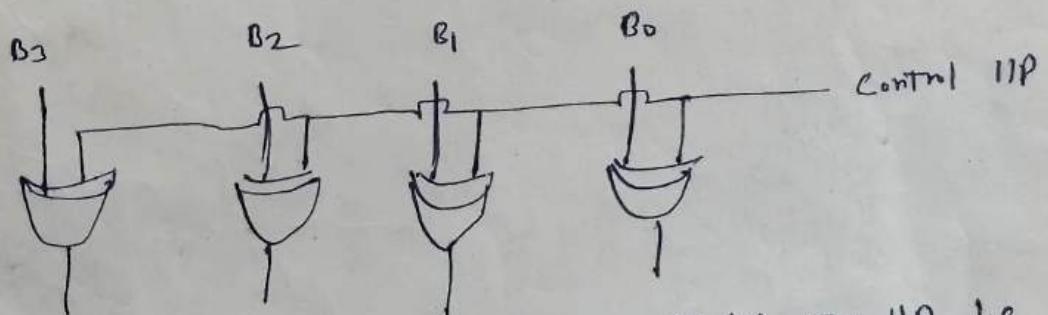
When one of the I/P is low the O/P is the Truth value of the other I/P & when one of the I/P is high the O/P is the complement of the other I/P.

X is the I/P
C control I/P

If $C = 0$ then $Y = X$
 $C = 1$

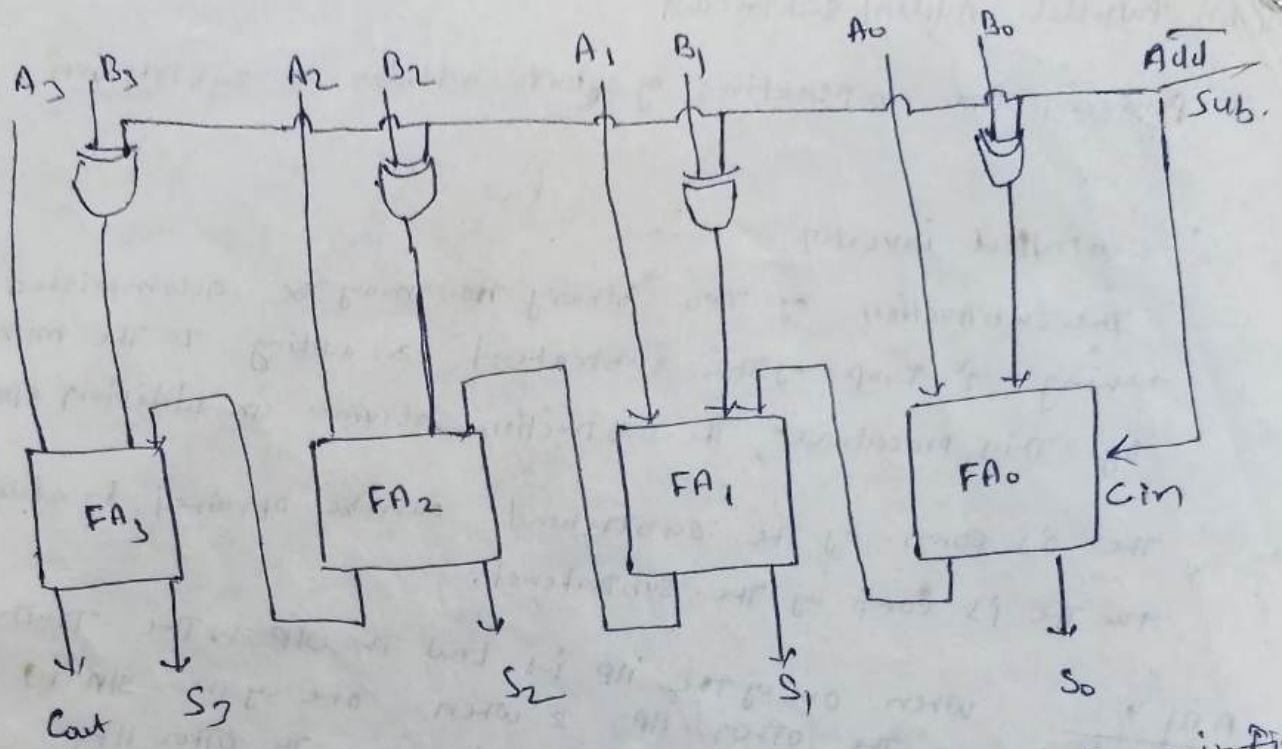
I/P X is available at Y in uncomplemented form.
I/P X is available at Y in complement form.

If $C = 1$, the EXOR gate can function as an inverter.



When the control I/P is low, the O/P will be the I/P i.e.

$B_3 B_2 B_1 B_0$, when the control I/P is high, the O/P will be the complement of the I/P $\bar{B}_3 \bar{B}_2 \bar{B}_1 \bar{B}_0$.



The 4-bit Parallel binary adder/subtractor performs the operations of both addition & subtraction. Ckt shown in fig.

$$\begin{array}{r}
 A_3 A_2 A_1 A_0 \\
 + \quad B_3 B_2 B_1 B_0 \\
 \hline
 \end{array}$$

1



Q. Simplify the expression

$$Y(A, B, D) = (\bar{A} + B)(A + B + D)\bar{D}$$

Given Boolean expression = $(\bar{A} + B)(A\bar{D} + B\bar{D} + D\bar{D})$
i.e.

$$= (\bar{A} + B)(A\bar{D} + B\bar{D})$$

Since $D \cdot \bar{D} = 0$

$$= \bar{A}A\bar{D} + \bar{A}B\bar{D} + AB\bar{D} + BB\bar{D}$$

Since $A\bar{A} = 0$

$B \cdot B = B$

$$= \bar{A}B\bar{D} + AB\bar{D} + B\bar{D}$$

$$= B\bar{D}(\bar{A} + A) + B\bar{D}$$

Since $\bar{A} + A = 1$

$$= B\bar{D} + B\bar{D}$$

$$= \underline{B\bar{D}}$$

Q. Simplify the following three-variable expression using Boolean algebra.

$$Y(A, B, C) = \sum m(0, 1, 3, 4, 7)$$

$$m_0 = 000 = \bar{A}\bar{B}\bar{C}$$

$$m_1 = 001 = \bar{A}\bar{B}C$$

$$m_3 = 011 = \bar{A}BC$$

$$m_4 = 100 = A\bar{B}\bar{C}$$

$$m_7 = 111 = ABC$$

SOP form

$$Y(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}\bar{B}(\bar{C} + C) + BC(\bar{A} + A) + A\bar{B}\bar{C}$$

$$= \bar{A}\bar{B} + BC + A\bar{B}\bar{C}$$

Q Simplify $y(A, B, C) = \sum m(0, 2, 4, 6)$

$$\begin{aligned} y(A, B, C) &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} \\ &= \bar{A}\bar{C}(B+B) + A\bar{C}(\bar{B}+B) \\ &= \bar{A}\bar{C} + A\bar{C} \\ &= \bar{C}(\bar{A}+A) \\ &= \bar{C} \text{ Ans} \end{aligned}$$

Q. Simplify the following

$$y(A, B, C) = \sum m(0, 1, 2, 3, 4, 5, 6, 7)$$

$$\begin{aligned} y(A, B, C) &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C \\ &\quad + AB\bar{C} + ABC \\ &= \bar{A}\bar{B}(\bar{C}+C) + \bar{A}B(\bar{C}+C) + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C) \\ &= \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB \\ &= \bar{A}(\bar{B}+B) + A(\bar{B}+B) \\ &= \bar{A} + A \\ &= 1 \text{ Ans} \end{aligned}$$

Q Simplify

$$y(A, B, C) = \sum m(1, 3, 5, 7)$$

$$m_1 = 001 = (A+B+\bar{C})$$

$$m_3 = 011 = (A+\bar{B}+\bar{C})$$

$$m_5 = 101 = (\bar{A}+B+\bar{C})$$

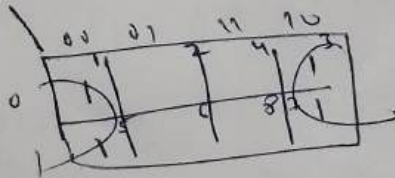
$$m_7 = 111 = (\bar{A}+\bar{B}+\bar{C})$$

$$Y(A,B,C) = (A+B+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$$

$$= \underline{\bar{A}BC}$$

Q. Minimize the expression using K-map

$$Y(A,B,C) = \sum m(0,1,4,5)$$



$$Y = \bar{B}$$

Q Minimize the following logic fun. using K-map

$$Y(A, B, C, D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$$

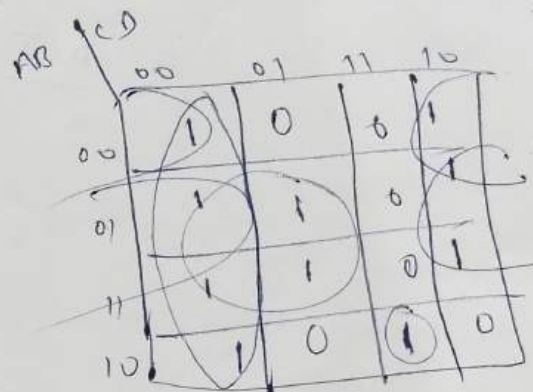
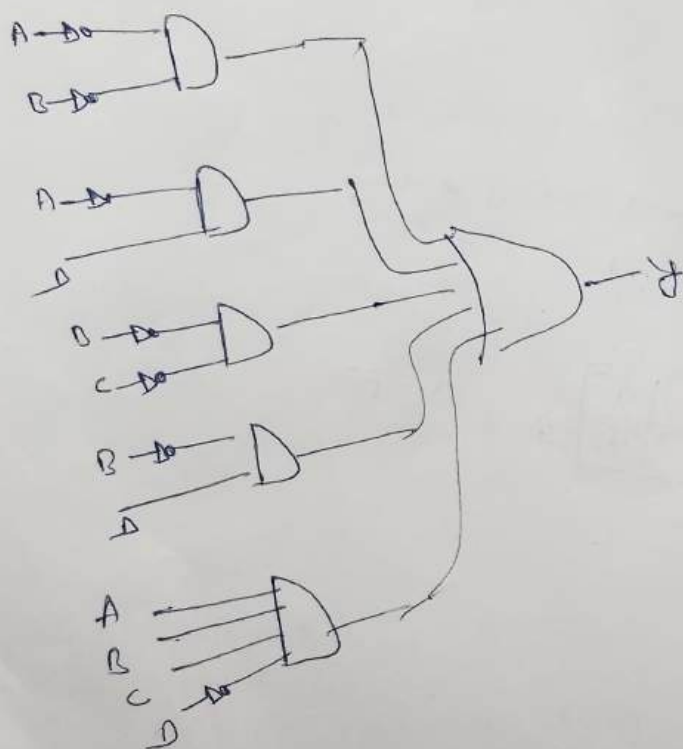
& implement it using logic gates.

Simplified Boolean expression is

$$Y(A, B, C, D) = \bar{A}\bar{B} + \bar{A}D + \bar{B}C + \bar{B}D + ABC\bar{D}$$

This expression can be implemented using

AND, OR & NOT gates.



Q. Minimize following Boolean expression using K-map

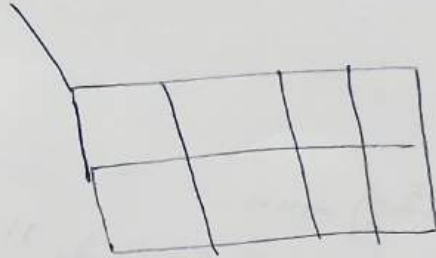
(a) $Y(ABC) = AB + C\bar{B} + C$

$$Y(ABC) = AB\bar{C} + BC + BC\bar{C}$$

$$Y(ABC) = AB(C + \bar{C}) + C\bar{B}(A + \bar{A}) + C(A + \bar{A})(B + \bar{B})$$

$$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}BC$$

$$Y(ABC) = \sum m(7, 6, 5, 1, 3)$$



$$= C + AB$$

$$= AB\bar{C} + A\bar{B}\bar{C} + ABC + \bar{A}BC + A\bar{B}C + \bar{A}B\bar{C}$$

(b) $Y(ABC) = \sum m(13, 12, 15, 7, 14, 6)$

$$= AB + BC$$

Q. Minimize the following Boolean expression using
K-map

(a) $Y(A,B,C,D) = AB + C\bar{B} + AC$
 $= ABC + BCD + B$

(b)

(a) $C\bar{B} + AB$ Ans

(b) $\bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D$ Ans

Minimize the following fun.

$Y(A,B,C,D) = \sum m(1,3,5,8,9,11,15) + d(2,13)$

$Y(A,B,C,D) = \prod M(1,2,3,8,9,10,11,14) \cdot d(7,15)$

(a) $\bar{B}D + \bar{C}D + A\bar{B}\bar{C} + AB$

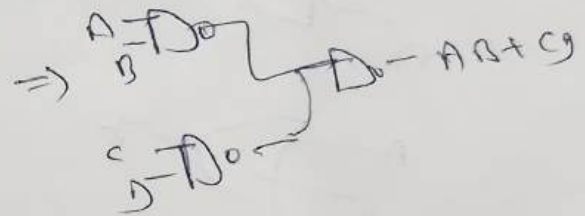
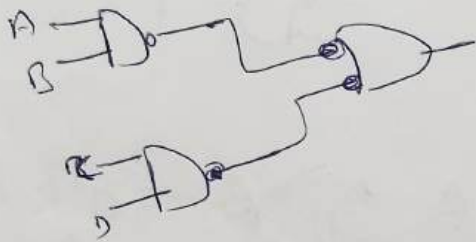
(b) $(\bar{A}+D)(\bar{A}+\bar{C})(B+\bar{C})(B+\bar{D})$

Q. $y = A\bar{B}C$, min no. of NAND gate required
 (a) 3 (b) 4 (c) 5 (d) 7

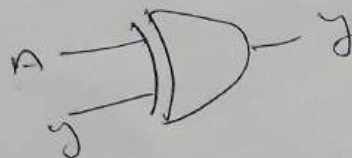
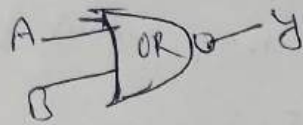
$$A\bar{B}C = AC\bar{B}$$

AND - 2 NAND
 NOT 1 NAND
 = 5

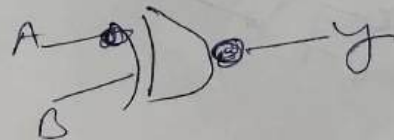
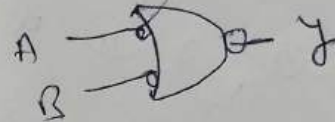
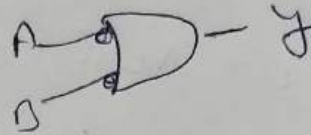
Q. to implement $AB + CD$ min no. of NAND gate
 (a) 3 (b) 4 (c) 4 (d) 5



LOGIC GATE

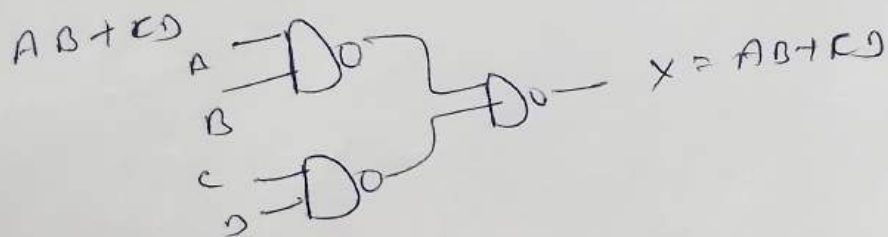
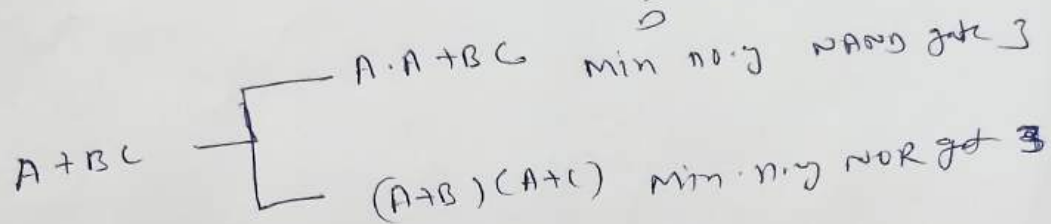
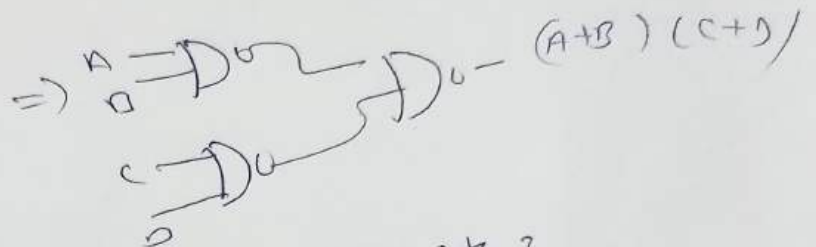
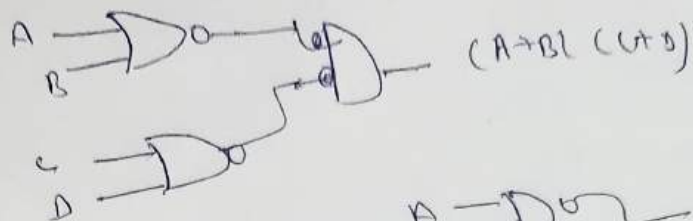


ALTERNATIVE Sy.



4.

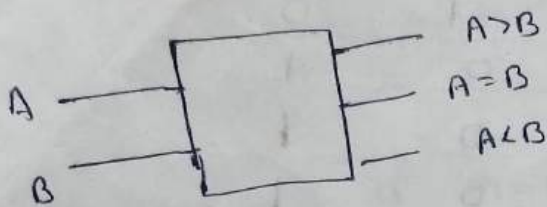
$$y = (A+B)(C+D)$$



Comparator

Comparator

1-bit comparator



$$\begin{aligned} A\bar{B} \\ \bar{A}\bar{B} + AB &= A \oplus B \\ \bar{A}B \end{aligned}$$

Truth table:-

AB	X A > B	Y A = B	Z A < B
00	0	1	0
01	0	0	1
10	1	0	0
11	0	1	0

Logic expressions:-

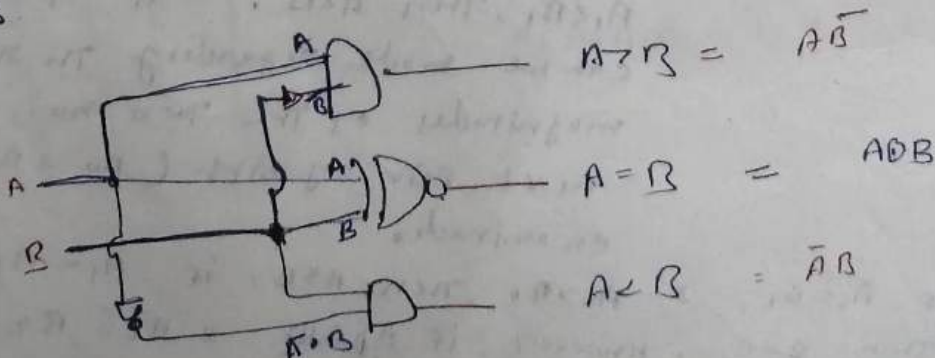
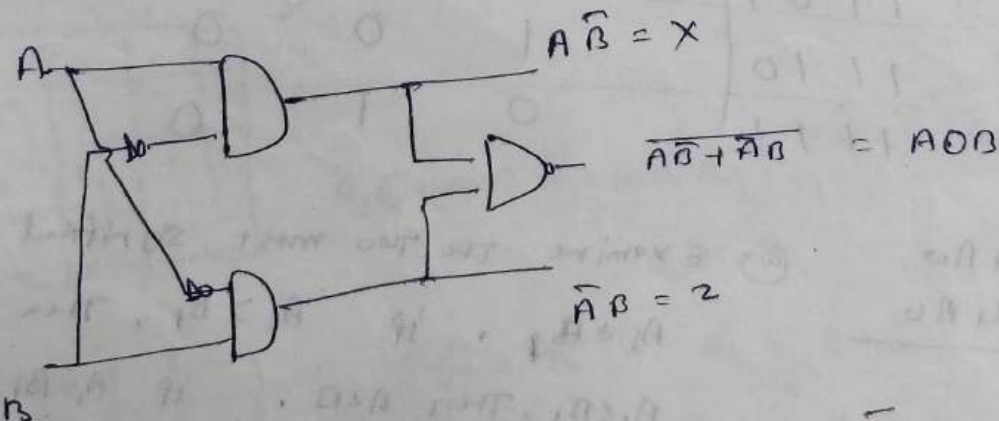
$$X = A\bar{B}$$

$$Y = \bar{A}\bar{B} + AB = A \oplus B$$

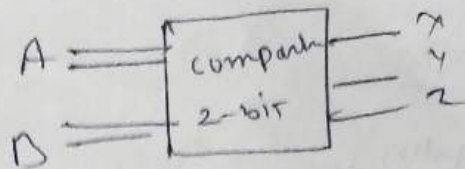
$$Z = \bar{A}B$$

Note For 1-bit comparator

1. 2 input
2. 4 combination
3. equivalence condition 2
4. greater than condⁿ 1
5. Lesser than condⁿ 1



2-bit comparator



A_1	A_0	B_1	B_0	$A > B$	$A = B$	$A < B$
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

$A_1 A_0$

$B_1 B_0$

(a) Examine the two most significant bit

$A_1 \neq B_1$, if $A_1 > B_1$, then $A > B$

$A_1 < B_1$, then $A < B$. If $A_1 = B_1$, no decision can be made regarding the relative magnitudes of the two no. & next pair of bits (A_0 & B_0) must be examined.

(b) if $A_1 = B_1$, & $A_0 > B_0$, then $A > B$, if $A_1 = B_1$, & $A_0 < B_0$, then $A < B$, however if $A_1 = B_1$ & $A_0 = B_0$, then $A = B$

if the most significant bit are equal
 (i.e. $A_1 = B_1 = 0$ or, $A_1 = B_1 = 1$) Then

$$E_1 = A_1 \odot B_1$$

if the next two most significant bit are equal,

$$E_0 = A_0 \odot B_0$$

if the hence if $A = B$, Then

$$E = E_1 E_0$$

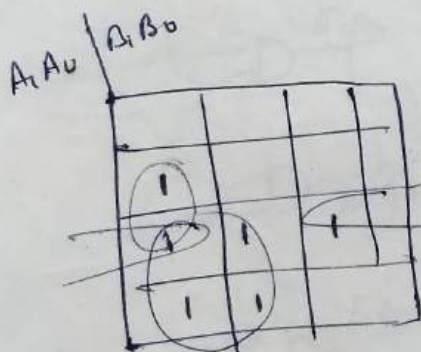
$$= (A_1 \odot B_1) (A_0 \odot B_0) = 1$$

$A > B$ is

$$A_1 \bar{B}_1 + E_1 A_0 \bar{B}_0$$

$$\begin{aligned} & (\bar{A}_1 \bar{B}_1 + A_0 B_0) A_0 \bar{B}_0 \\ & \bar{A}_1 \bar{B}_1 A_0 \bar{B}_0 + \bar{A}_0 \bar{B}_0 A_0 \bar{B}_0 \\ & \bar{A}_1 \bar{B}_1 A_0 \bar{B}_0 + A_0 \end{aligned}$$

$$A < B = \bar{A}_1 B_1 + E_1 \bar{A}_0 B_0$$



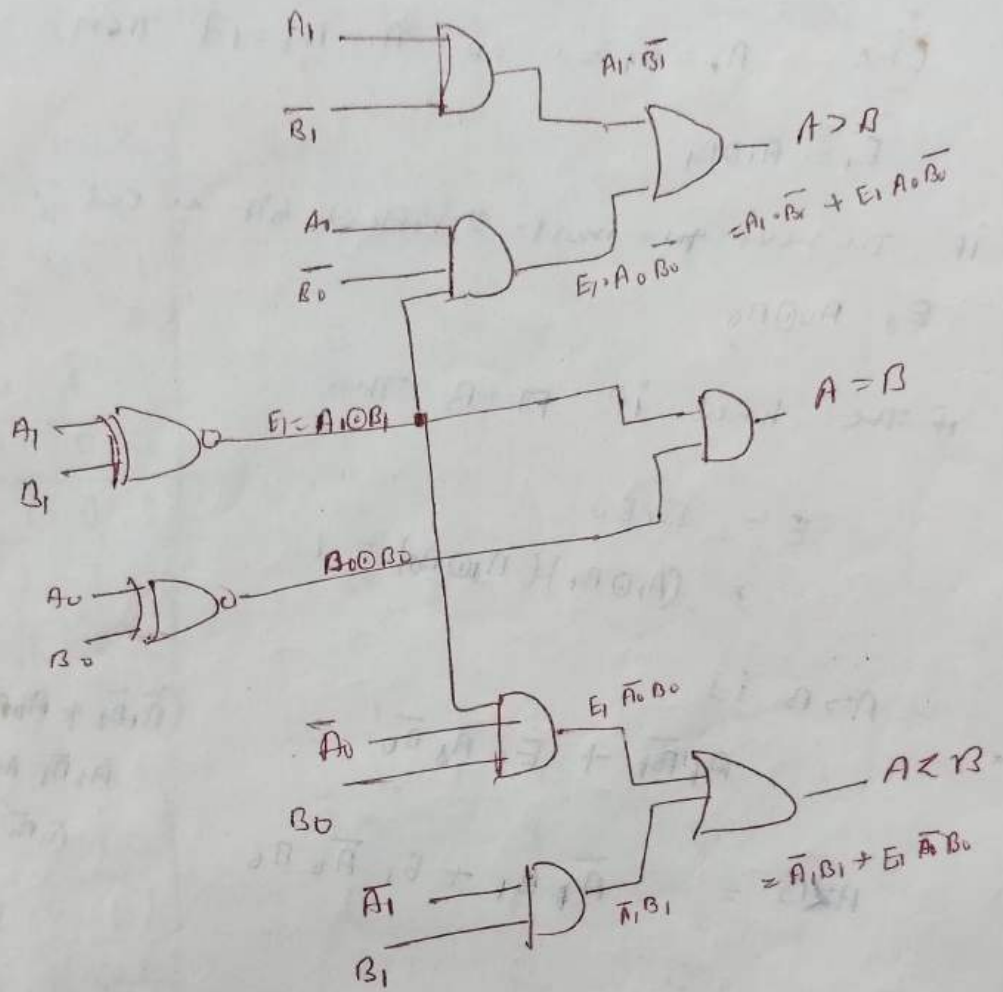
$$A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$



$$\bar{A}_1 \bar{B}_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_1 \bar{B}_0$$

$$A_1 A_0 B_1 B_0 + \bar{A}_1 A_0 \bar{B}_1 B_0$$

$$\bar{A}_1 B_1 + \bar{A}_1 \bar{B}_0 + A_0 + \bar{A}_1 B_1 B_0$$



Four-bit magnitude comparator →

$$A = A_3 A_2 A_1 A_0$$

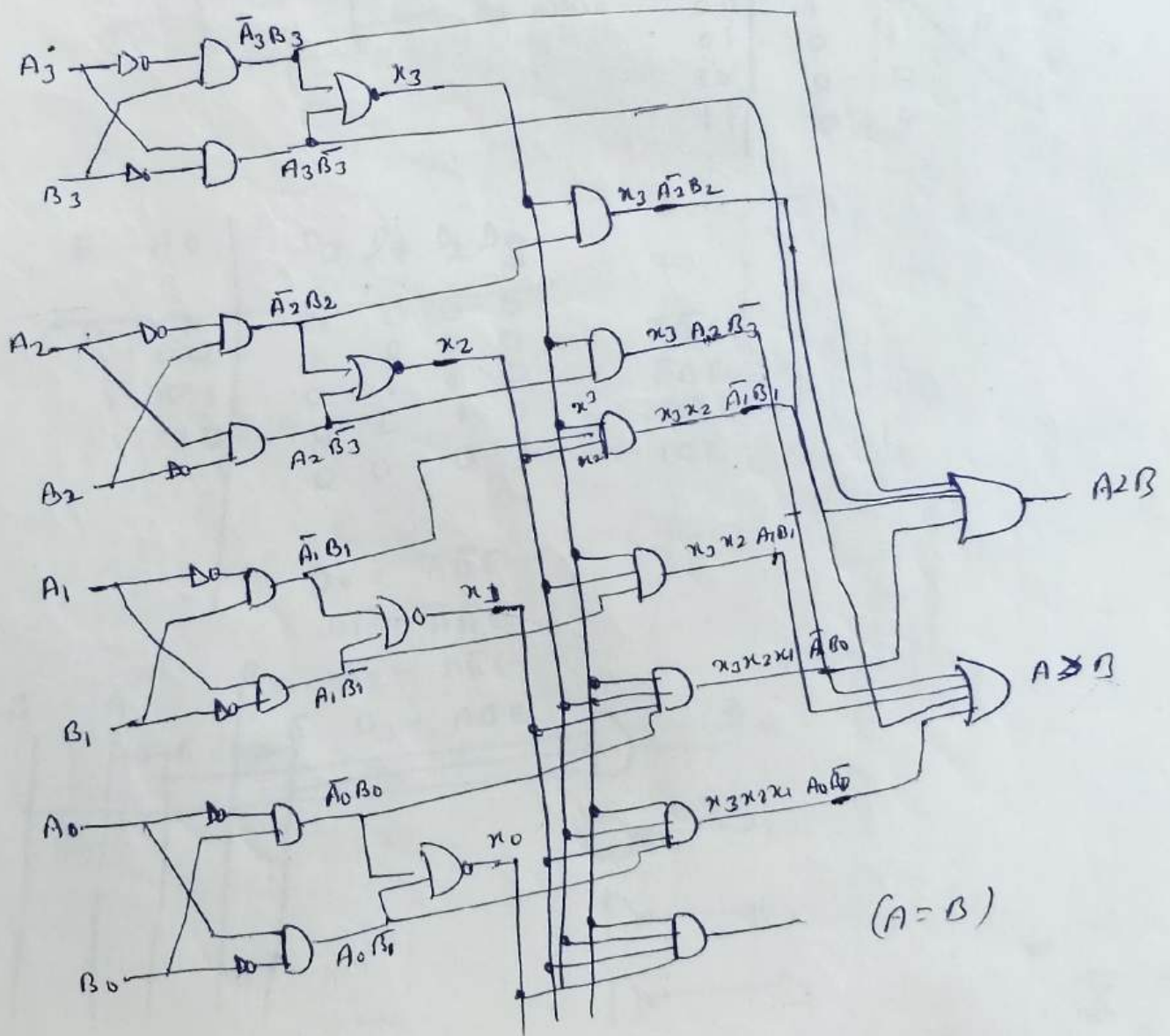
$$B = B_3 B_2 B_1 B_0$$

The two nos are equal if all pairs of significant digits are equal $A_3=B_3, A_2=B_2, A_1=B_1, A_0=B_0$

$$x_i = A_i B_i + \bar{A}_i \bar{B}_i =$$

$$A > B \quad A_3 \bar{B}_3 + x_3 A_2 \bar{B}_2 + x_3 x_2 A_1 \bar{B}_1 + x_3 x_2 x_1 A_0 \bar{B}_0$$

$$A < B \quad \bar{A}_3 B_3 + x_3 \bar{A}_2 B_2 + x_3 x_2 \bar{A}_1 B_1 + x_3 x_2 x_1 \bar{A}_0 B_0$$



Classification of codes →

The digital data as an input to machine contains alphabets, numbers, punctuations, & special characters. These should be represented by unique combinations of 1 & 0. These combinations of 1 & 0 are referred as the code for info. When the data is transmitted over a long distance, it is transmitted in terms of code-words. To detect & correct the errors, special codes are used in digital sy. comm.

The commonly used binary codes are classified as:

- ① Weighted code
- ② Self-complementary code.
- ③ Unit-distance codes.
- ④ Alphanumeric codes.
- ⑤ Cyclic codes
- ⑥ Error detecting & correcting codes.

Weighted code

The weight of a digit or a bit depends on its position

decimal code - 589
The weight of 5 is 500
 8 is 80
 9 is 9

similar binary code,

Binary, BCD, 8-4-2-1 & 2-4-2-1 are examples of weighted codes.

Self complementary code -

Excess-3 code is the example of self-comp. The 1's comp. of the Excess 3-code is the Excess-3 code for the 9's complement of the corresponding decimal no.

Excess-3 code for 2 is 0101,
1's comp of 0101 is 1010.

It is the excess-3 code of 7 & 7 is the 9's comp of 2.

unit distance codes →

The name itself indicates that there is unit distance b/w two consecutive codes. That is, the bit patterns for two consecutive codes differ in only one bit position.

Gray code

Alphanumerical codes The binary codes of alphabets, numbers & special symbols are known as alphanumerical codes.

ASCII American standard code for inf. interchange.

Error detecting & correcting codes —

When inf. in digital form is transmitted to a long dist, errors may get introduced & 1 becomes 0 & 0 becomes 1. Special codes are used to detect & correct such errors. Parity & Hamming codes are commonly used for error detecting & correction.

Gray code →

Gray code is a 4-bit numeric code. decimal no, 0 to 15, are represented by 4-bit binary codes. This code is also known as unit distance code because two consecutive codes differ in only one bit position.

Gray code is an unweighted code, the bit positions in the code group do not have any specific weight assigned to them. Bcoz of this, the Gray code is not suitable for arithmetic operations, but it find application in

analog to digital converters.

Binary to Gray code converters →

Decimal digit	Binary B ₃ B ₂ B ₁ B ₀	Gray. G ₃ G ₂ G ₁ G ₀
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

→ difference

$\begin{matrix} 0010 \\ \downarrow \\ 0011 \end{matrix}$
 $\begin{matrix} 0101 \\ \downarrow \\ 0111 \end{matrix}$

one bit of the Gray code changes for two consecutive digits
 ex. if the decimal digit changes from 6 to 7, then
 Gray code from 0101 to 0100! There is changes in
 leftmost bit only. This is the principle characteristic of
 the Gray code.

Binary $G_3 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$

$G_2 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$

$G_1 = \Sigma m(2, 3, 4, 5, 10, 11, 12, 13)$

$G_0 = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14)$

$G_3 =$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$G_3 = B_3$

$G_2 =$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

$G_2 = \bar{B}_3 B_2 + B_3 \bar{B}_2$

$G_2 = B_3 \oplus B_2$

$G_1 =$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1

$G_1 = B_2 \bar{B}_1 + \bar{B}_2 B_1$

$G_1 = B_2 \oplus B_1$

$G_0 =$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$G_0 = \bar{B}_1 B_0 + B_1 \bar{B}_0$

$G_0 = B_1 \oplus B_0$

then
the
digits

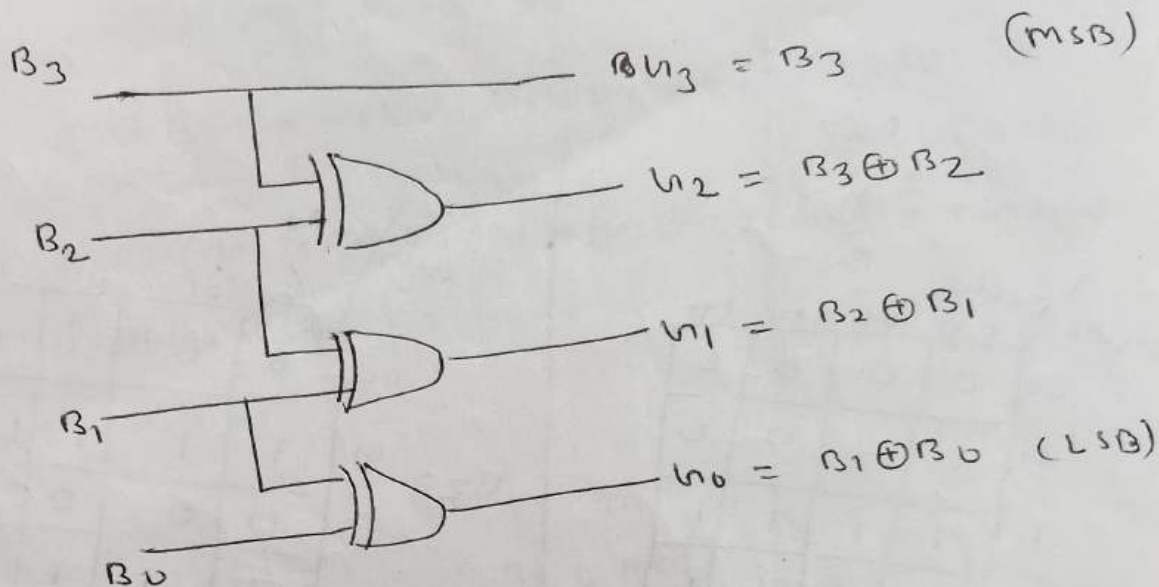
$B \rightarrow W$

① $w_3 = B_3$

$w_2 = B_3 \oplus B_2$

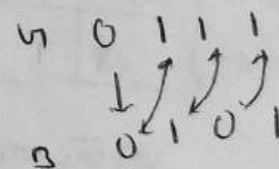
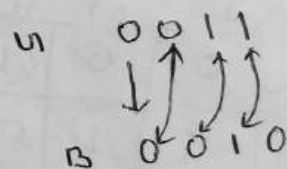
$w_1 = B_2 \oplus B_1$

$w_0 = B_1 \oplus B_0$



Gray code to Binary converters $\rightarrow$

w_3	w_2	w_1	w_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0



$$B_3 = \Sigma m(8, 9, 10, 11, 12, 13, 14, 15)$$

$$B_2 = \Sigma m(4, 5, 6, 7, 8, 9, 10, 11)$$

$$B_1 = \Sigma m(2, 3, 4, 5, 8, 9, 14, 15)$$

Draw ~~K-map~~ K-map.

$$B_3 = h_3$$

$$B_2 =$$

$$B_1 =$$

$$B_0 =$$

B_3

$h_3 h_2$ \ $h_1 h_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$B_3 = h_3$$

B_1

$h_3 h_2$ \ $h_1 h_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

$$B_1 = \overline{h_3} \overline{h_2} h_1 + \overline{h_3} h_2 \overline{h_1} + h_3 h_2 h_1 + \overline{h_3} h_2 \overline{h_1}$$

$$= \overline{h_3} (\overline{h_2} h_1 + h_2 \overline{h_1}) + h_3 (h_2 h_1 + h_2 \overline{h_1})$$

$$= \overline{h_3} (h_2 \oplus h_1) + h_3 (h_2 \oplus h_1)$$

$$= h_3 \oplus h_2 \oplus h_1$$

$$B_1 =$$

$$B_1 = B_2 \oplus h_1$$

B_2

$h_3 h_2$ \ $h_1 h_0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

$$B_2 = \overline{h_3} h_2 + h_3 \overline{h_2}$$

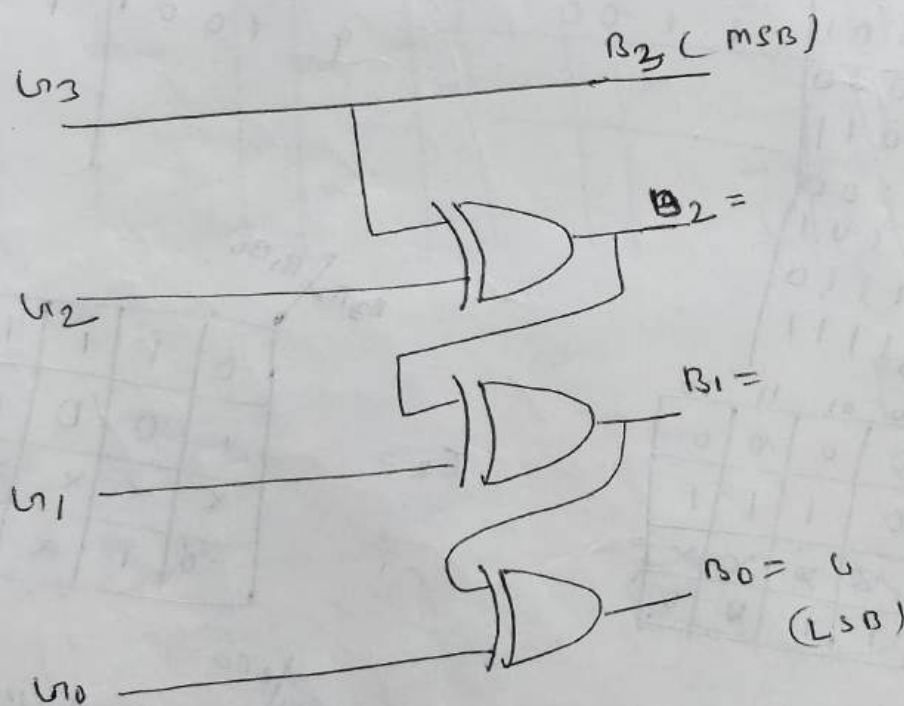
$$B_2 = h_3 \oplus h_2$$

B_0

$h_3 h_2$ \ $h_1 h_0$	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

$B_0 =$

$$\begin{aligned}
 B_0 &= \bar{u}_3 \bar{u}_2 \bar{u}_1 u_0 + \bar{u}_3 \bar{u}_2 u_1 \bar{u}_0 + u_3 u_2 \bar{u}_1 u_0 + u_3 u_2 u_1 \bar{u}_0 \\
 &\quad + \bar{u}_3 u_2 \bar{u}_1 \bar{u}_0 + u_3 \bar{u}_2 \bar{u}_1 \bar{u}_0 + \bar{u}_3 u_2 u_1 u_0 + u_3 \bar{u}_2 u_1 u_0 \\
 &= \bar{u}_3 \bar{u}_2 (\bar{u}_1 u_0 + u_1 \bar{u}_0) + u_3 u_2 (\bar{u}_1 u_0 + u_1 \bar{u}_0) + \bar{u}_1 \bar{u}_0 (u_3 u_2 + u_3 \bar{u}_2) \\
 &\quad + u_1 u_0 (\bar{u}_3 u_2 + u_3 \bar{u}_2) \\
 &= \bar{u}_3 \bar{u}_2 (\bar{u}_0 \oplus u_1) + u_3 u_2 (u_0 \oplus u_1) + \bar{u}_1 \bar{u}_0 (u_2 \oplus u_3) + u_1 u_0 (u_2 \oplus u_3) \\
 &= (u_0 \oplus u_1) (\bar{u}_3 \bar{u}_2 + u_3 u_2) + (u_2 \oplus u_3) (\bar{u}_1 \bar{u}_0 + u_1 u_0) \\
 &= u_0 \oplus u_2 \oplus u_1 \oplus u_3 \\
 B_0 &= u_0 \oplus B_1
 \end{aligned}$$



BCD TO Excess-3 converter

Excess-3 code is a modified BCD code. It is obtained by adding 3 to each BCD code.

BCD code $B_3 B_2 B_1 B_0$	$E_3 E_2 E_1 E_0$
0 0000	0 0 1 1
1 0001	0 1 0 0
2 0010	0 1 0 1
3 0011	0 1 1 0
4 0100	0 1 1 1
5 0101	1 0 0 0
6 0110	1 0 0 1
7 0111	1 0 1 0
8 1000	1 0 1 1
9 1001	1 1 0 0

BCD code $B_3 B_2 B_1 B_0$	Excess-3 $E_3 E_2 E_1 E_0$
0 0 0 0	0 0 1 1
1 0 0 0 1	0 1 0 0
2 0 0 1 0	0 1 0 1
3 0 0 1 1	0 1 1 0
4 0 1 0 0	0 1 1 1
5 0 1 0 1	1 0 0 0
6 0 1 1 0	1 0 0 1
7 0 1 1 1	1 0 1 0
8 1 0 0 0	1 0 1 1
9 1 0 0 1	1 1 0 0

$E_3 =$

$B_3 B_2$ / $B_1 B_0$	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$E_2 =$

$B_3 B_2$ / $B_1 B_0$	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

$E_1 =$

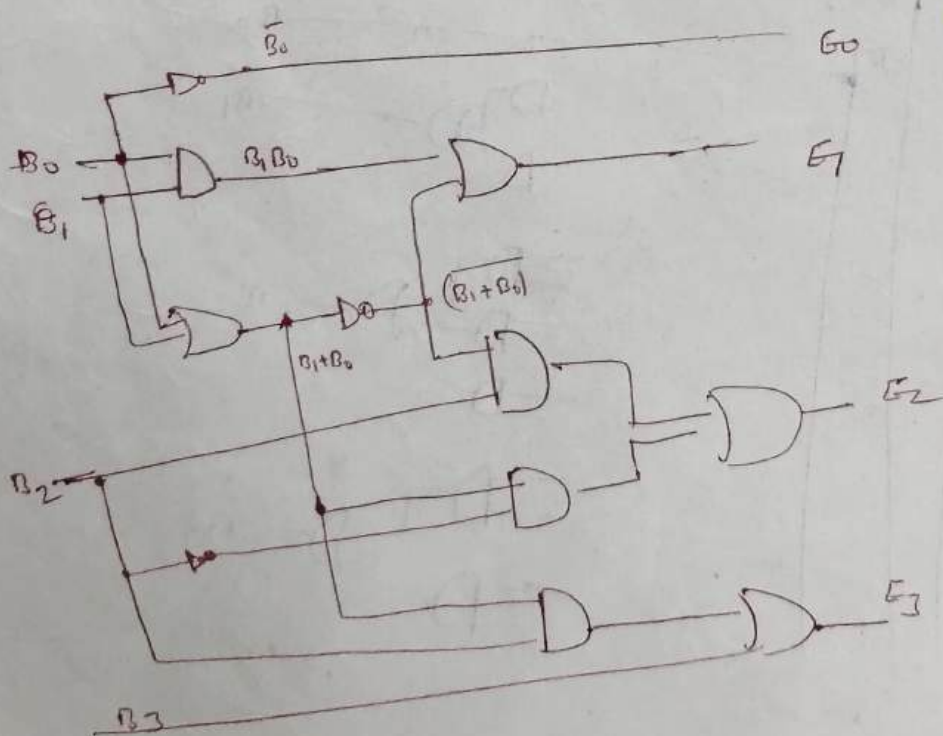
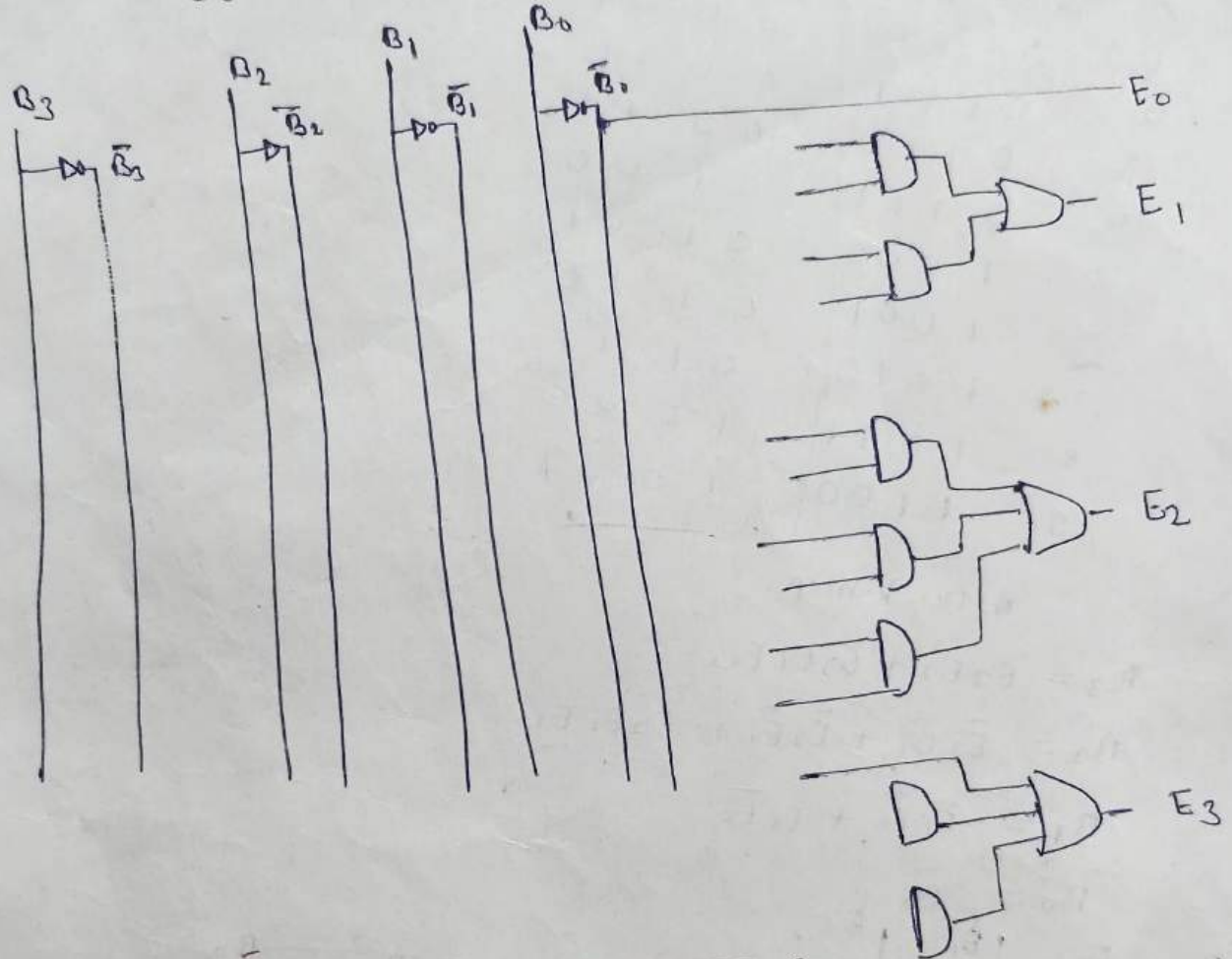
$B_3 B_2$ / $B_1 B_0$	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

$E_0 =$

$B_3 B_2$ / $B_1 B_0$	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

$$E_0 = \overline{B_0}$$

$$\begin{aligned}
 E_3 &= B_3 + B_2 B_0 + B_2 B_1 = B_3 + B_2 (B_0 + B_1) \\
 E_2 &= \bar{B}_2 B_0 + \bar{B}_2 B_1 + B_2 \bar{B}_1 \bar{B}_0 = \bar{B}_2 (B_0 + B_1) + B_2 \bar{B}_1 \bar{B}_0 \\
 E_1 &= \bar{B}_1 \bar{B}_0 + B_1 B_0 = B_1 B_0 + (\bar{B}_1 + \bar{B}_0) \\
 E_0 &= \bar{B}_0
 \end{aligned}$$



Excess-3 To BCD converter

decimal digit	<u>Excess-3</u>				<u>BCD</u>			
	E_3	E_2	E_1	E_0	B_3	B_2	B_1	B_0
0	0	0	1	1	0	0	0	0
1	0	1	0	0	0	0	0	1
2	0	1	0	1	0	0	1	0
3	0	1	1	0	0	0	1	1
4	0	1	1	1	0	1	0	0
5	1	0	0	0	0	1	0	1
6	1	0	0	1	0	1	1	0
7	1	0	1	0	0	1	1	1
8	1	0	1	1	1	0	0	0
9	1	1	0	0	1	0	0	1

After K-map.

$$B_3 = E_3 E_2 + E_3 E_1 E_0$$

$$B_2 = \bar{E}_2 \bar{E}_1 + \bar{E}_2 \bar{E}_0 + E_2 E_1 E_0$$

$$B_1 = \bar{E}_1 \bar{E}_0 + E_1 \bar{E}_0$$

$$B_0 = \bar{E}_0$$

